

Managerial Decision Making
Business 38802 (AXP 25)
Autumn 2025
George Wu

Class 4

Risky Decision Making

Course Outline

Monday, November 3	Introduction to Decision Making
Tuesday, November 4	Judgment under Uncertainty I (Heuristics and biases)
Wednesday, November 5*	Judgment under Uncertainty II (The difficulty of learning)
Friday, November 7	Risky Decision Making
Saturday, November 8	Nudges/Improving Decision Processes
January 5	Final (More next)

Sample Final Question

Consider the following fact: approximately 50% of small businesses fail within their first three years of operations. A friend is unaware of this fact and wishes to quit his or her job as an investment banker and open a restaurant in Barcelona. As his or her good friend, you want to be sure overconfidence is not playing a significant role in his or her decision. As you consider how to discuss this issue with your friend, consider the following questions:

- a. Why might your friend be overconfident about the success of this venture? Please be specific in identifying the most important sources of overconfidence.
- b. Which of these sources do you believe will be most resistant to your persuasion?

Final

■ Logistics

- ❑ In class, January 5, 2026 (!)
- ❑ 2 hours
- ❑ 3 out of 5 questions
- ❑ 600 words per question
- ❑ Open book (but Chat GPT and other AI **strictly** prohibited!)

■ Samples

- ❑ Sample Question “answer”, more sample questions, and tips in handouts on Canvas

What did we learn last class?

- Feedback is necessary but not sufficient
 - In a nutshell, experience alone is not a prescription
 - Feedback is often incomplete
 - What would've happened if you did something different?
 -  Are there ways to get feedback that you wouldn't otherwise get?
 - Even if feedback is abundant, feedback is
 - Ambiguous
 - Registered asymmetrically
 - Success = skill
 - Failure = bad luck
 - Delayed/scattered over time
 - People/organizations aren't ready to use

What did we learn last class?

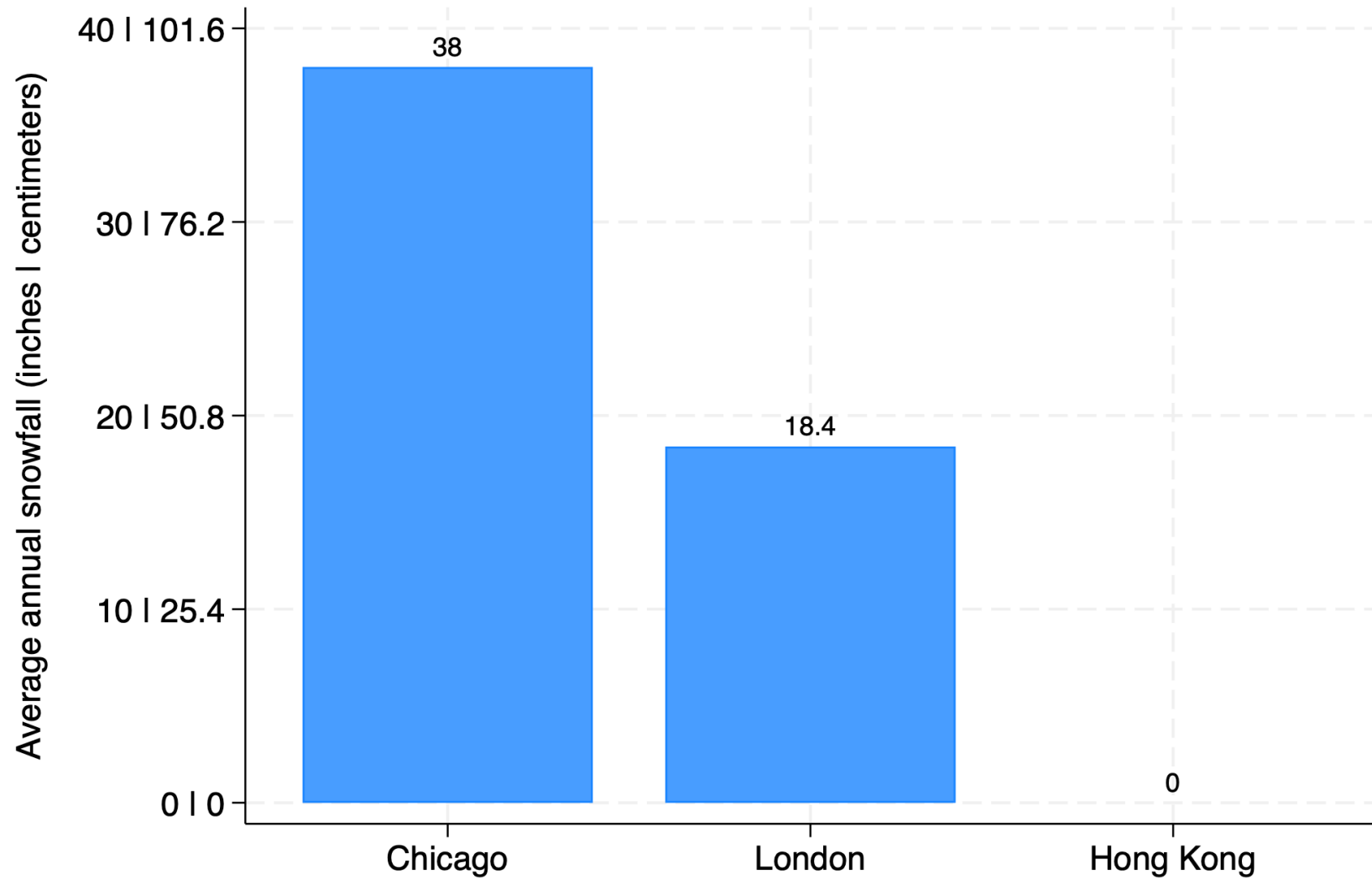
- Narratives rule! ... but
 - What's psychologically surprising is not statistically surprising (e.g., regression to the mean is almost impossible to understand intuitively)
 - We have great facility to spin stories to explain the facts ... but do not consider alternative stories that could also explain the data
 - UK football
 - Wason task
- Psychological biases
 - Confirmation bias
 - Hindsight bias

Today's Objectives

- Goals: Risky Decision Making
 - Descriptive model of risky decision making (prospect theory)
 - Mental Accounting

- Outline
 - First half: Application to
 - Toro
 - Second half: lecture on prospect theory and mental accounting
 - Application to recruiting and retention

Snowfall and XP



Snow in Hong Kong? City facing big chill but internet rumours could be flaky

Hong Kong Observatory says temperatures will plummet to single digits

Reading Time: 2 minutes

Why you can trust SCMP



Ernest Kao

Published: 10:15pm, 18 Jan 2016 | Updated: 1:38pm, 22 Jan 2016

Get ready for the big chill next week when temperatures will plunge to single digits – but don't get excited by unofficial weather forecasts predicting snow, the Hong Kong Observatory says.

It was quick to dismiss internet speculation of 1cm of snow this weekend, clarifying instead that a cold surge expected to reach southern China on Thursday would trigger a chilly spell.

J.P.Morgan PRIVATE BANK

When markets shift,
we stand strong.

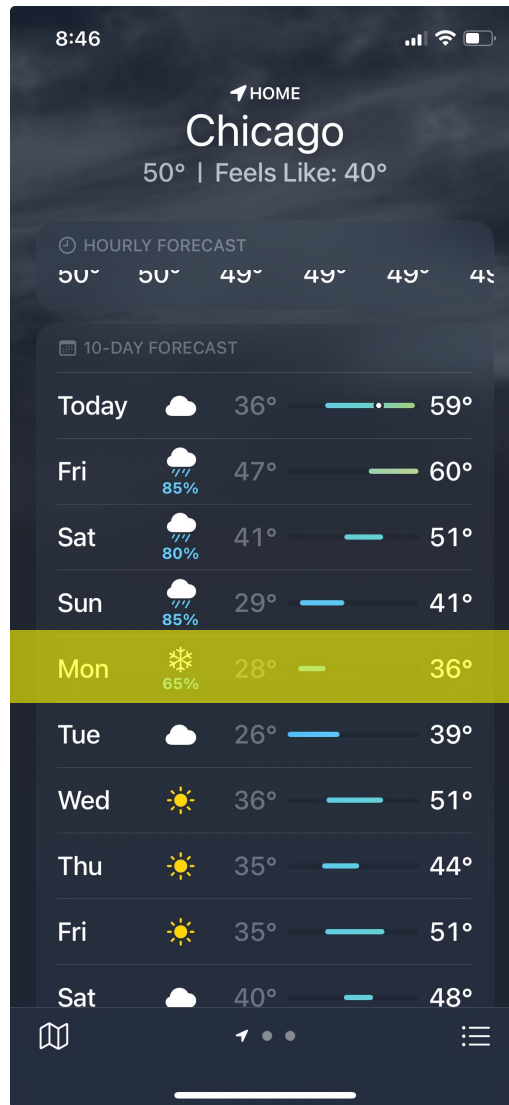
Lean on over 200 years
of strength and stability.

CONTACT US



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When Nick and I get back to Chicago...



Toro

NEW TORO MONEY BACK S'NO RISK PROGRAM

**IF IT DOESN'T SNOW WE'LL RETURN YOUR DOUGH.
AND YOU KEEP THE SNOWTHROWER!**



Buy a Toro Snowthrower before December 10, 1983, and if it snows less than 20% of average this season, you get all your money back and you keep the Toro. If it snows less than 50% of average you get 50% of your money back. And you still keep the Toro.

If it snows more than that you'll really be glad you bought a Toro snowthrower.

Offer includes S-140, S-200R, S-200E, S-620E, and all two stage models.

Come in for full details.

TORO

Haven't you done
without a Toro
long enough?[®]



(DEALER IMPRINT)

The S'No Risk Promotion

**IF IT DOESN'T
SNOW WE'LL
RETURN YOUR
DOUGH! AND
YOU KEEP THE
SNOWTHROWER.**



If it snows
less than

20%
* AVERAGE
SNOWFALL

30%
* AVERAGE
SNOWFALL

40%
* AVERAGE
SNOWFALL

50%
* AVERAGE
SNOWFALL

You keep the Toro and
you receive:

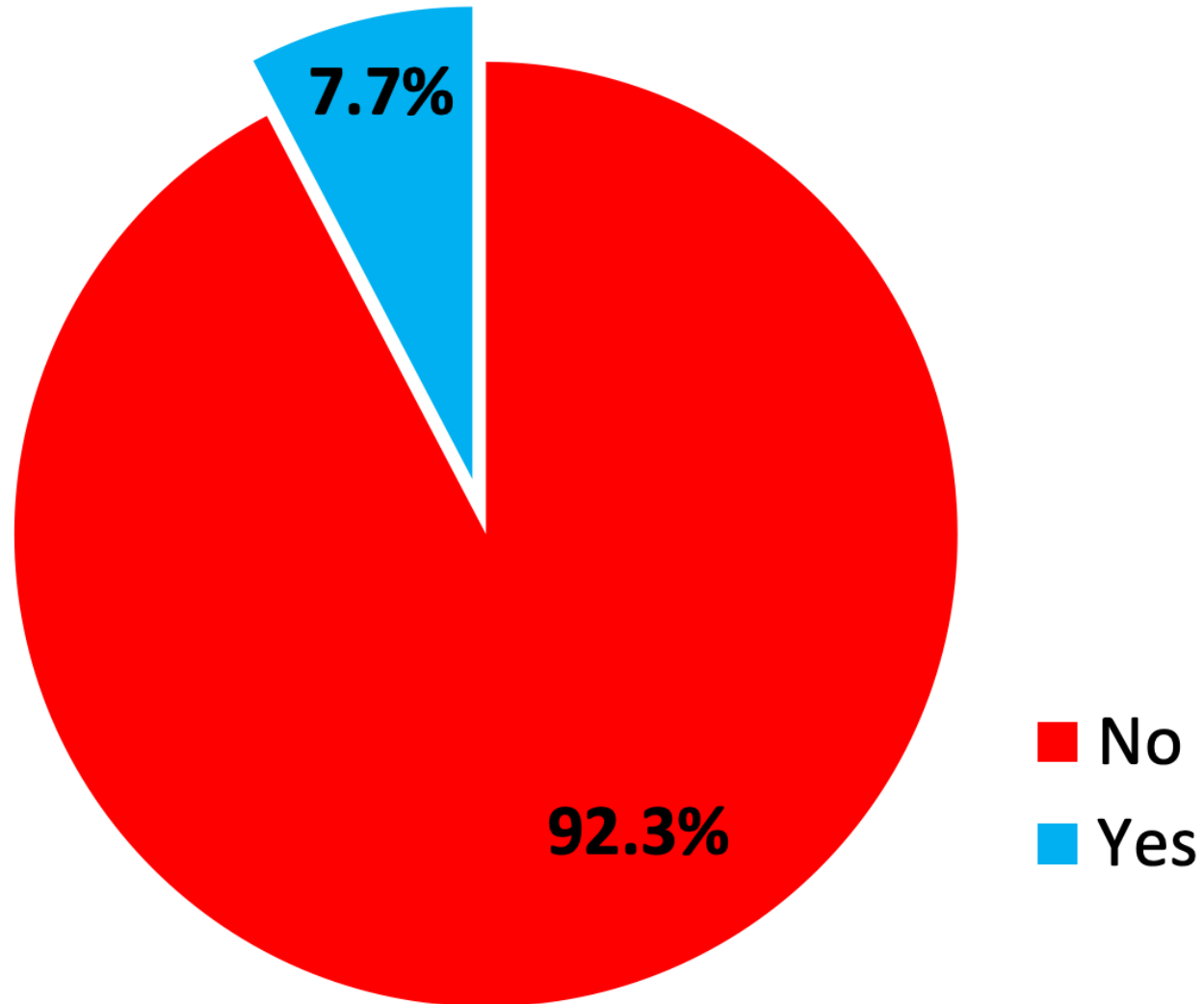
100%
REFUND
of suggested retail price

70%
REFUND
of suggested retail price

60%
REFUND
of suggested retail price

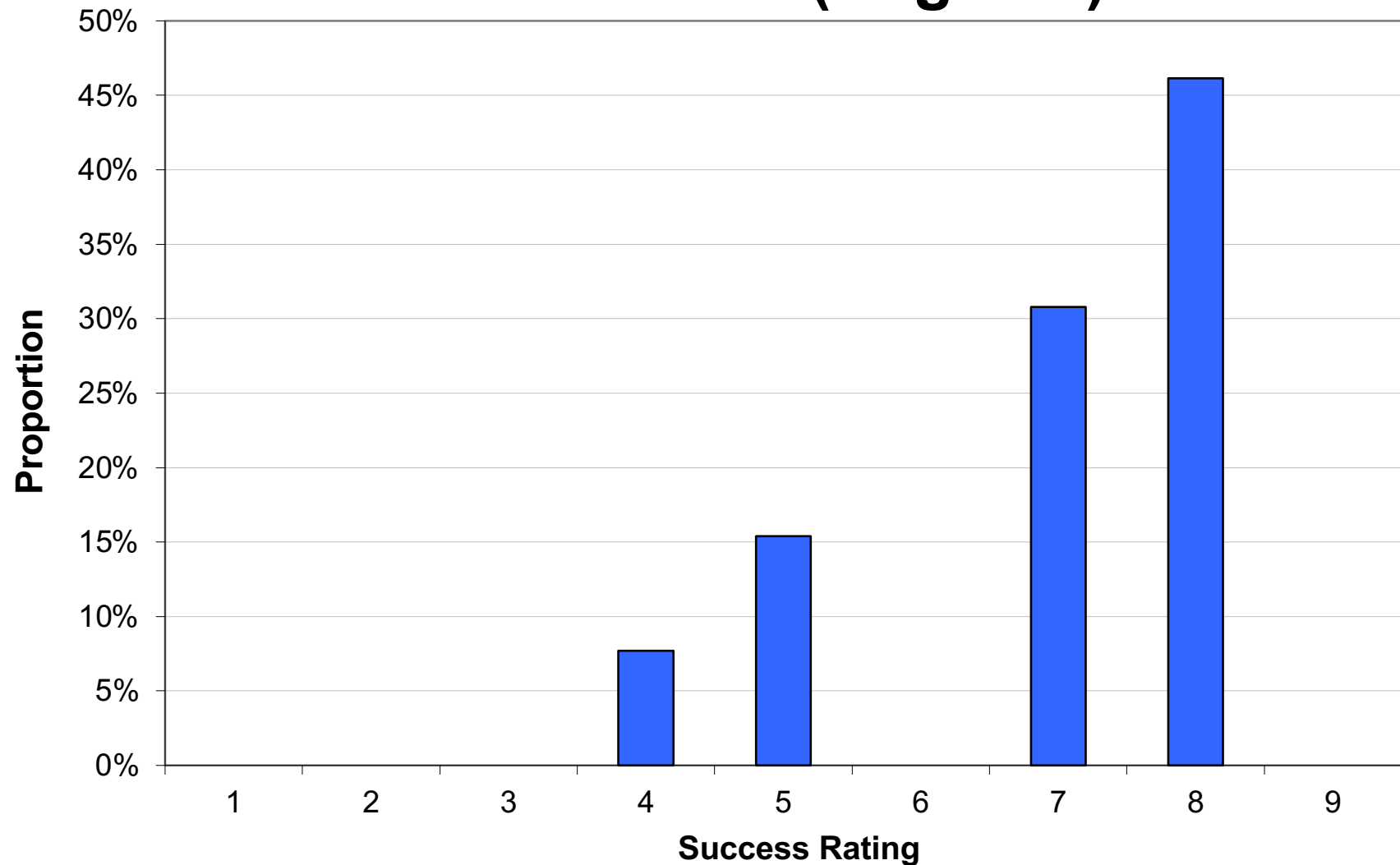
50%
REFUND
of suggested retail price

Should Toro repeat the program?



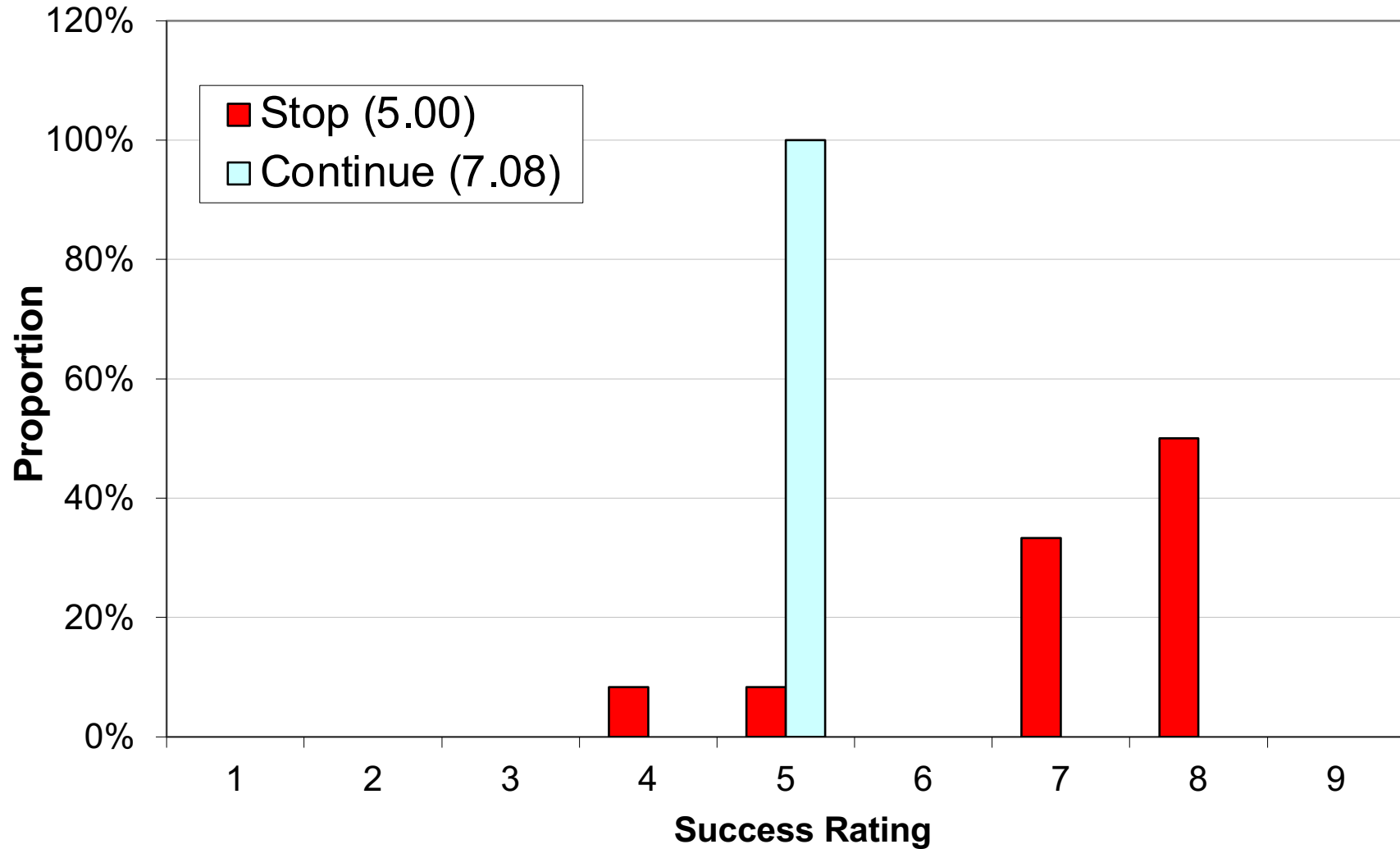
Was the S'No Risk program a success?

Toro Success (Avg: 6.9)



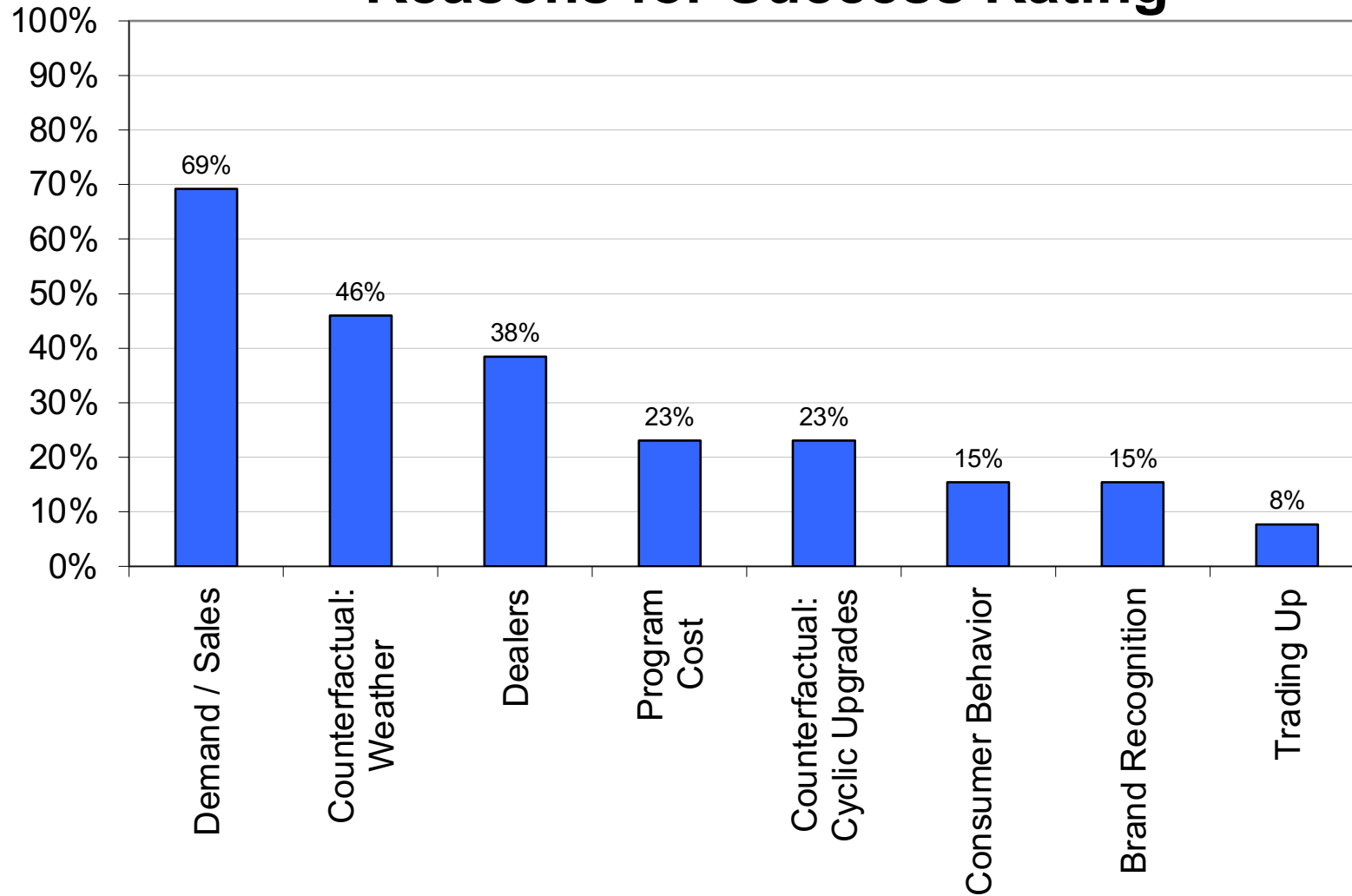
Was the S'No Risk program a success?

Toro Success



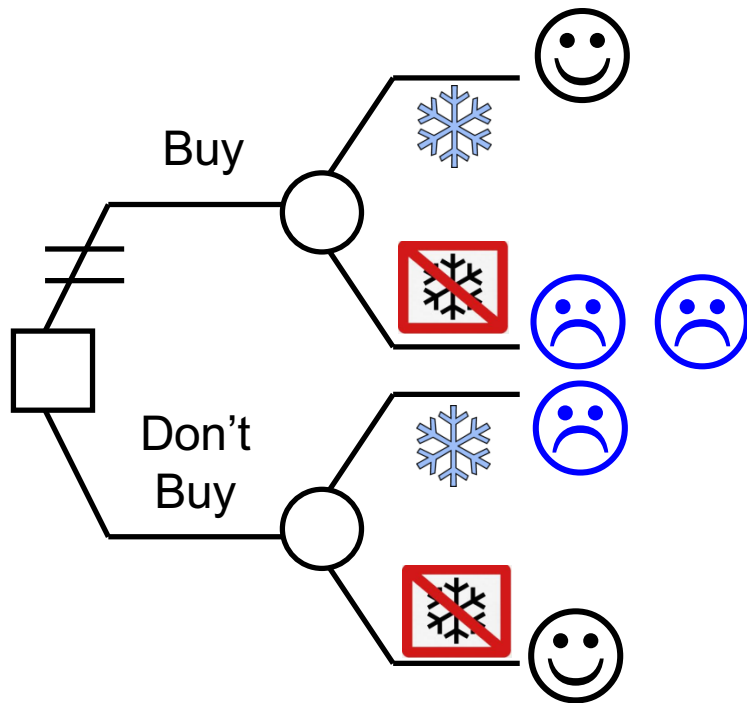
Provide a brief justification

Reasons for Success Rating

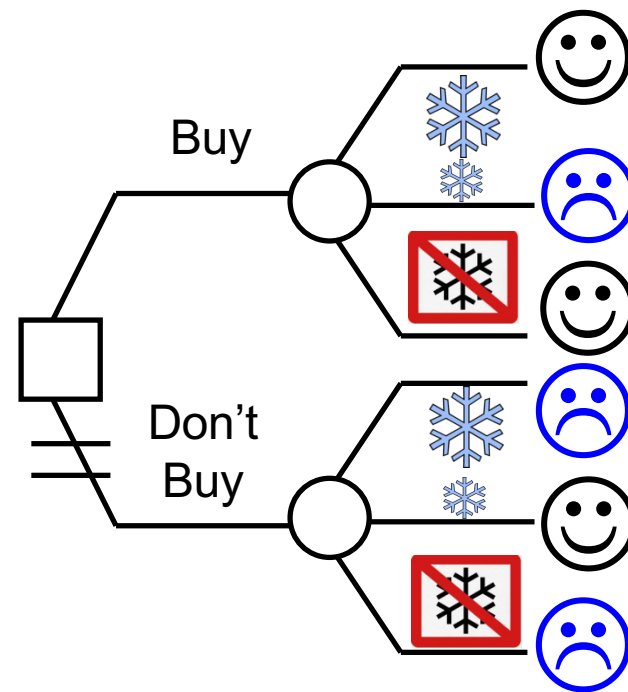


S'No Risk eliminates Regret

Without S'No Risk



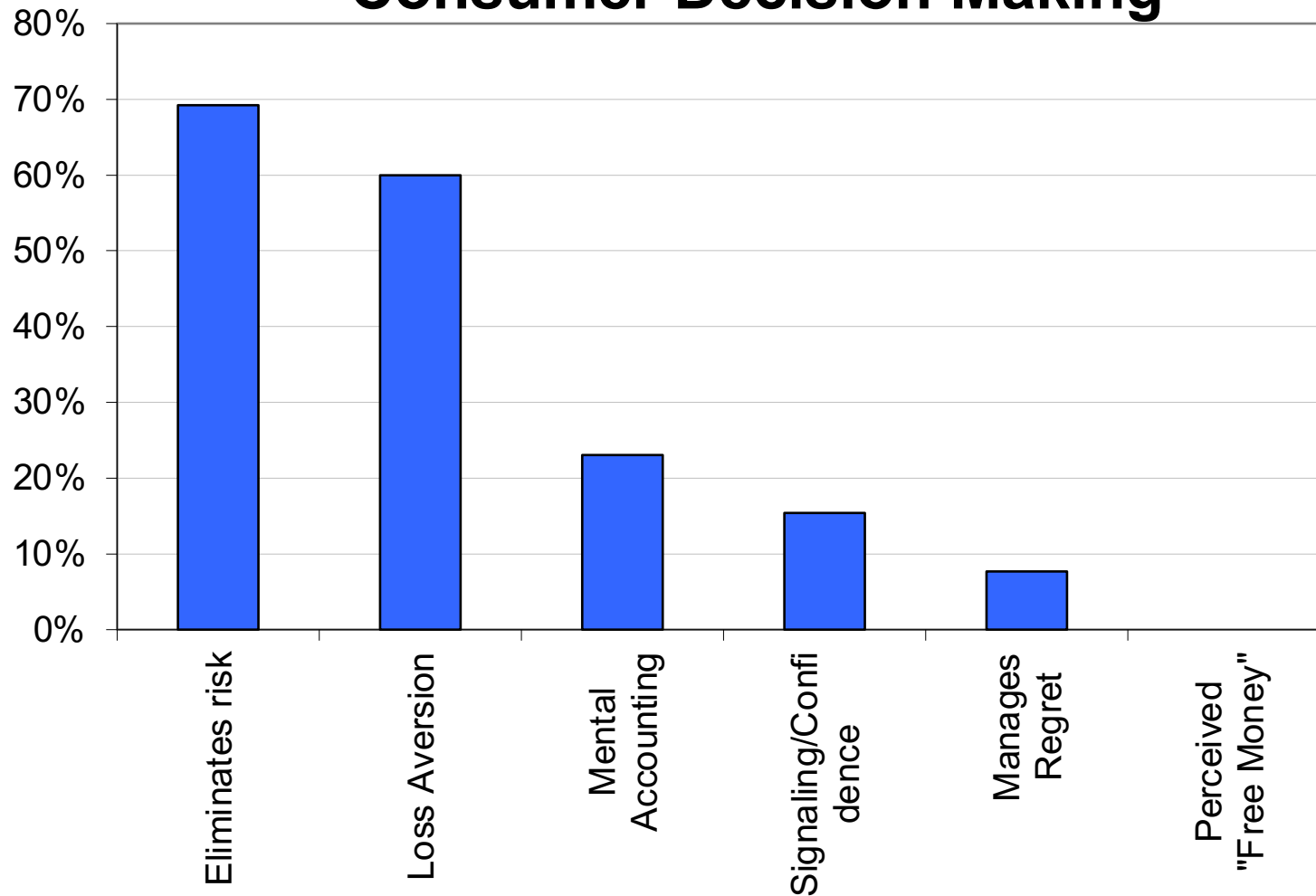
With S'No Risk



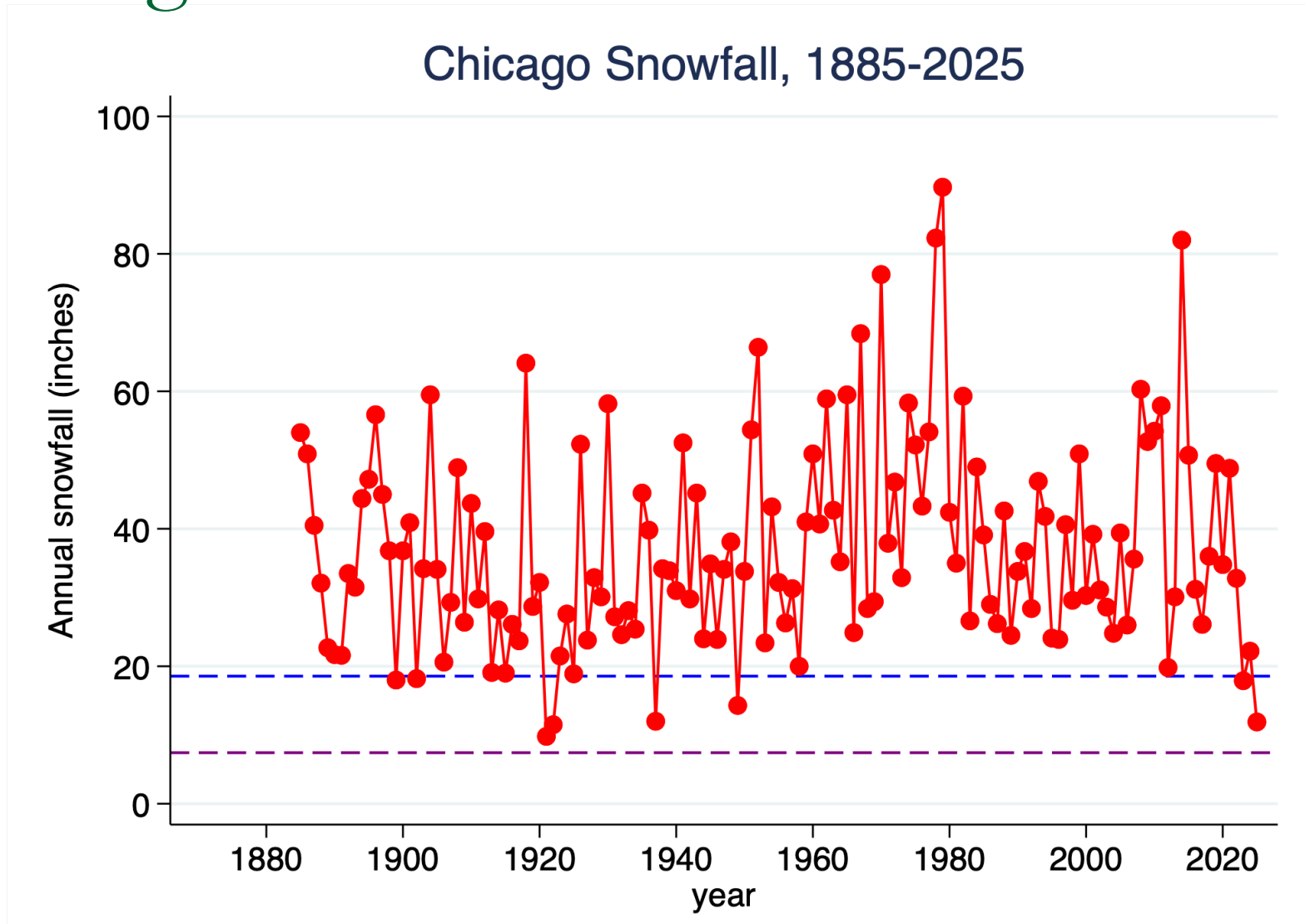
Where's the "sleight-of-hand"?

S'No Risk and consumer decisions

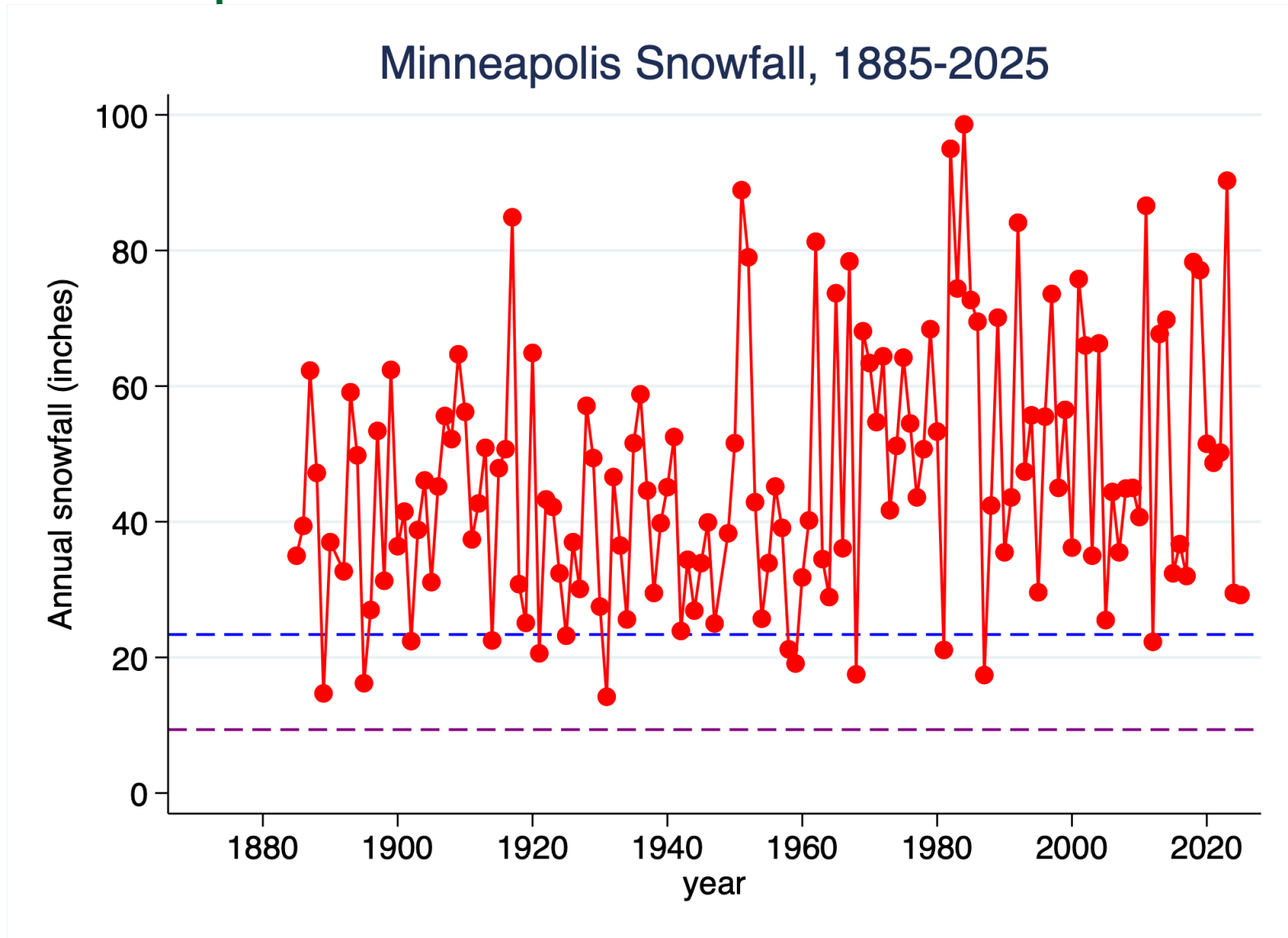
Consumer Decision Making



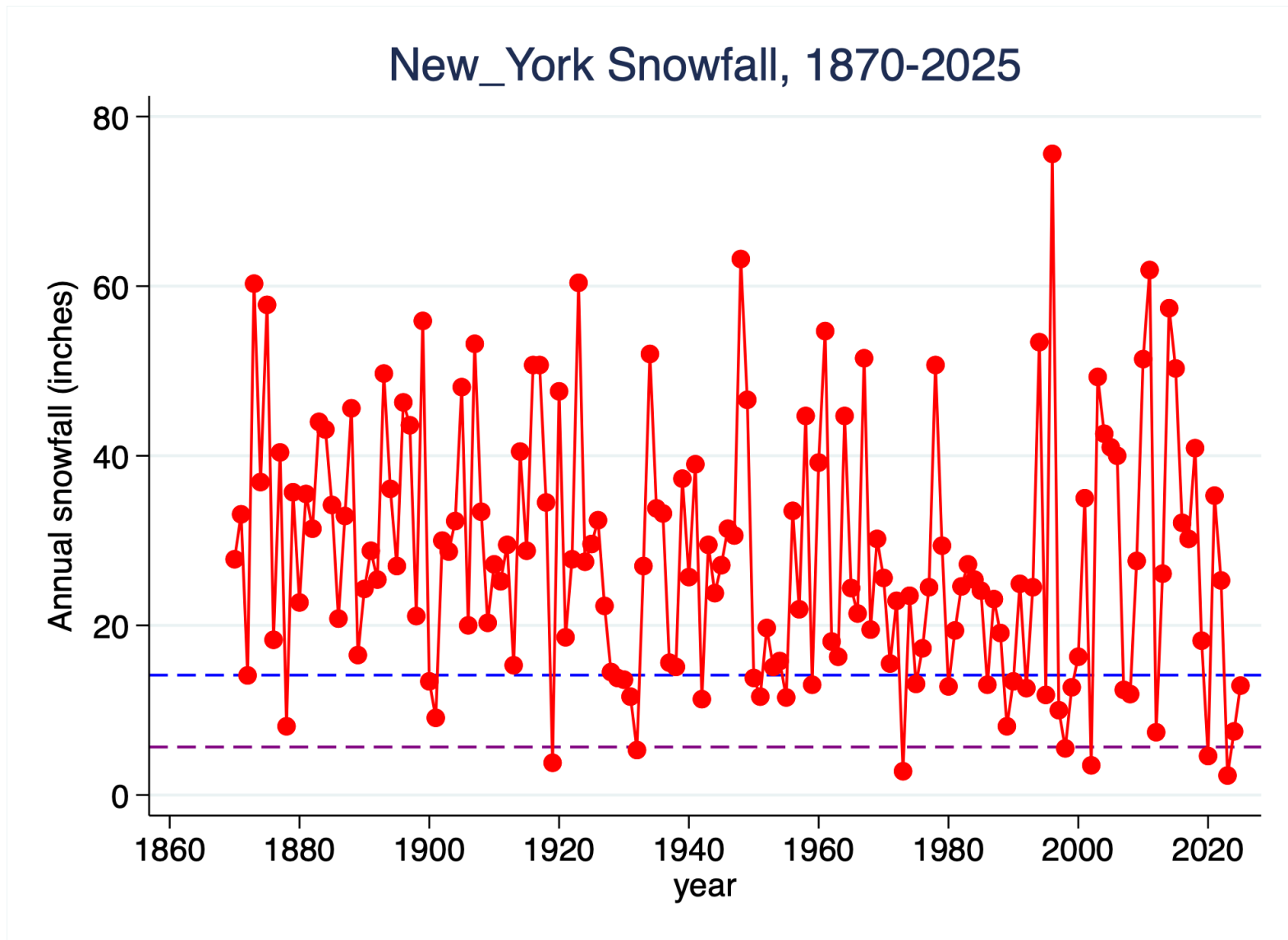
Chicago Annual Snowfall



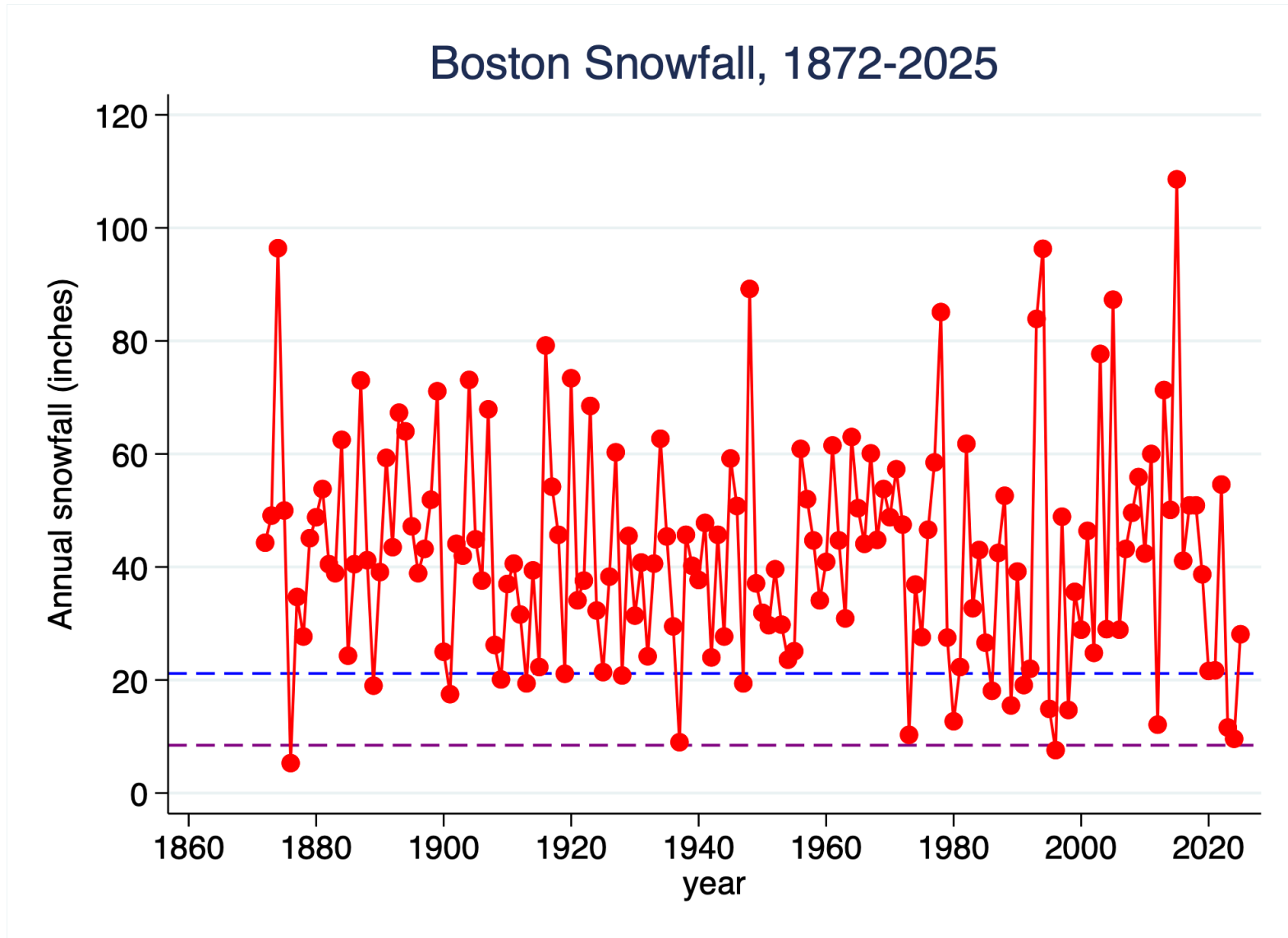
Minneapolis Annual Snowfall



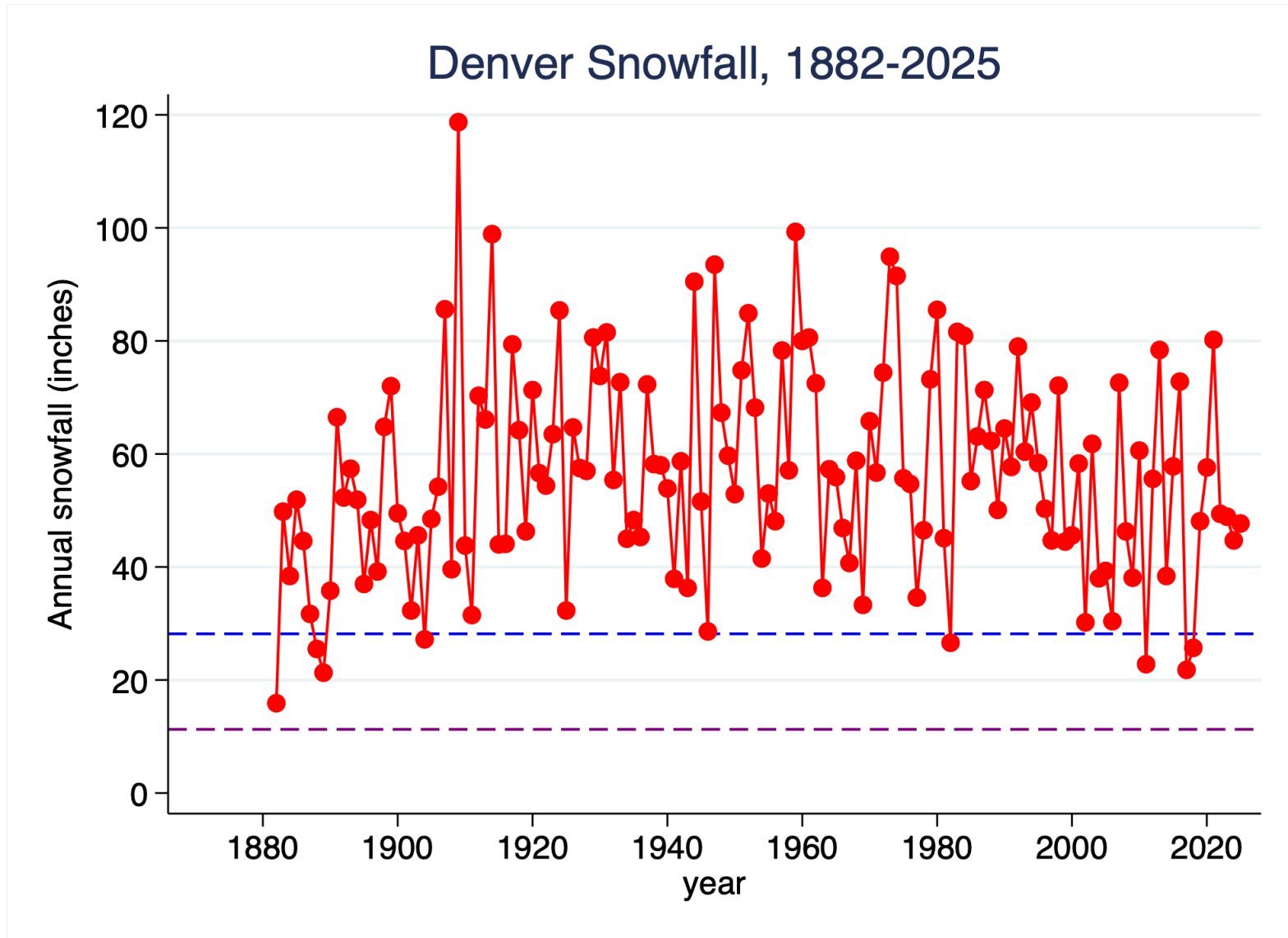
New York Annual Snowfall



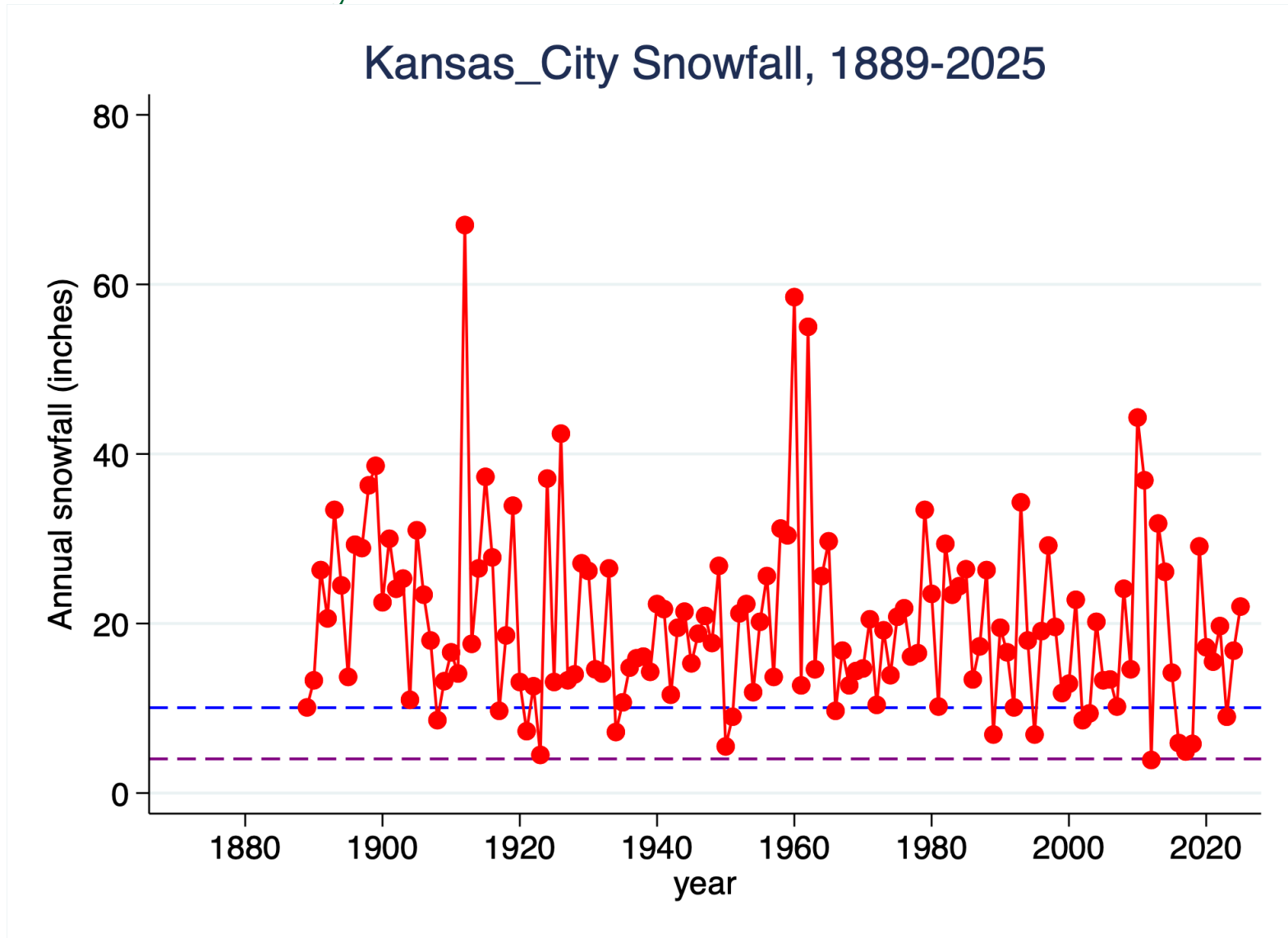
Boston Annual Snowfall



Denver Annual Snowfall



Kansas City Annual Snowfall



Chance of Collecting-Revised?

City	20% (100%)	30% (70%)	40% (60%)	50% (50%)	Prob Win	Exp Cost
Boston (136)	1.5%	4.4%	5.9%	12.5%	24.3%	14.3%
Chicago (138)	0.0%	0.7%	2.9%	4.4%	8.0%	4.4%
Denver (141)	0.0%	0.7%	2.1%	5.7%	8.5%	4.6%
Kansas City (134)	0.8%	4.5%	7.5%	13.4%	26.1%	15.1%
Minneapolis (136)	0.0%	0.0%	3.7%	9.6%	13.2%	7.0%
New York (153)	3.9%	5.9%	<u>7.8%</u>	19.6%	37.2%	22.5%

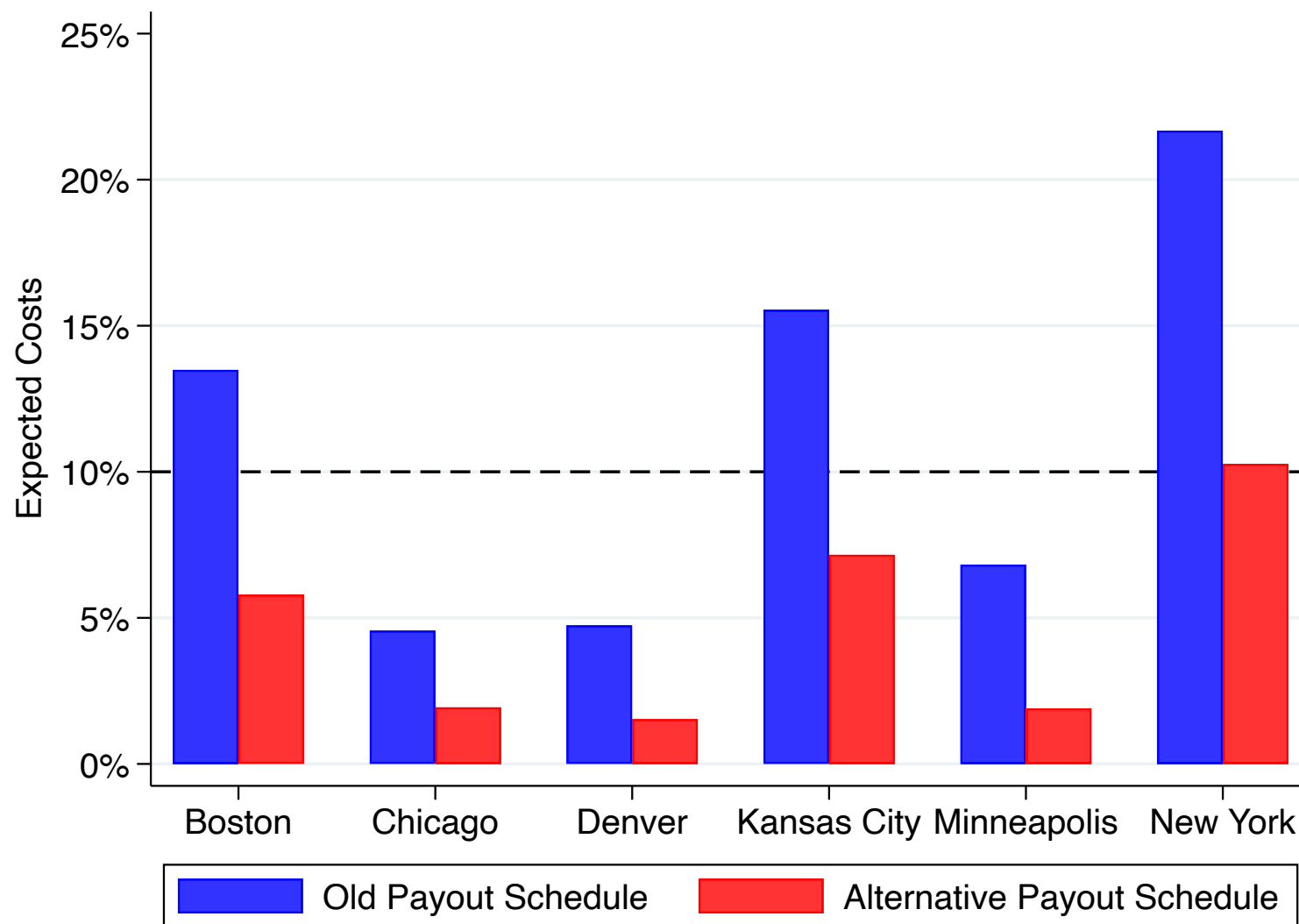
Improving Promotion?

IF IT DOESN'T SNOW WE'LL RETURN YOUR DOUGH! AND YOU KEEP THE SNOWTHROWER. 	20% * AVERAGE SNOWFALL	100% REFUND of suggested retail price
	30% * AVERAGE SNOWFALL	70% REFUND of suggested retail price
	40% * AVERAGE SNOWFALL	60% REFUND of suggested retail price
	50% * AVERAGE SNOWFALL	50% REFUND of suggested retail price

Chance of Collecting-Alternative?

City	10% (100%)	20% (70%)	30% (60%)	40% (50%)	Prob Win	Exp Cost
Boston (136)	0.0%	1.5%	4.4%	5.9%	11.8%	4.6%
Chicago (138)	0.0%	0.0%	0.7%	2.9%	3.6%	1.9%
Denver (141)	0.0%	0.0%	0.7%	2.1%	2.8%	1.5%
Kansas City (134)	0.0%	0.8%	4.5%	7.5%	12.7%	6.9%
Minneapolis (136)	0.0%	0.0%	0.0%	3.7%	3.7%	1.8%
New York (153)	0.7%	3.9%	5.9%	7.8%	18.3%	10.8%

Chance of Collecting-Revised?



What Happened

- Toro decided not to offer the S'No Risk program in 1984-85.
- However, they made the program available to individual distributors, at the distributor's option.

What Happened: One Distributor

- One Connecticut distributor offered the S'No Risk Program. They sold about \$2.5 million worth of snowblowers and paid out \$310,000 in claims to Hartford area residents [50% of purchase price (900 people got refunds)]
- The distributor did insure with Goodweather. They paid a fraction of sales as a premium to Goodweather, and would have received a 50% refund credit if there had been no claims.

What Happened: One Distributor

- The distributor's president said it was a great program. Despite the high insurance cost (since they didn't get the refund) they got lots of free publicity (TV news, etc.) and expected to get some additional advantage in the future: ("you didn't buy last year? Listen to Hartford resident Mr. Smith....")
- The president also saw that a major value of the promotion was in encouraging customers to buy bigger machines than they otherwise would have.

Weather Derivatives

WSJ, 2/2/99

Has Mother Nature Met Her Match?

Weather Derivatives Are Becoming a Popular Hedge

By MARK GOLDEN and ED SILLIERE
Dow Jones Newswires

Bombardier Inc.'s sales of snowmobiles, as with other seasonal items, have long been at the mercy of Mother Nature: A cold winter with early snowfall means a quick sellout; a mild winter means slashed prices and leftover inventory.

But this winter Bombardier beat the weather. The Montreal transportation-equipment maker offered buyers in 16 Midwestern cities a \$1,000 rebate on its Ski-Do snowmobiles, which marks a 10% to 33% cut based on the model, if snowfall in their areas didn't reach at least half the average of the past three years. To hedge its bet, it turned to snowfall options.

Under the snowfall options, one of a host of burgeoning weather derivatives, the company paid **Enron Corp.** \$45 to \$400 for each snowmobile sold, depending on the city. The Houston energy giant, in turn, agreed to reimburse Bombardier the full \$1,000 for every rebate paid.

Ski-Do sales in the 16 cities soared 38% from a year ago. Snowfall already has

weather through mid-December kept holiday shoppers out of their stores; then, blizzards and ice storms hamstrung big chunks of the U.S. just before Christmas. **Nautica Enterprises Inc.**, for example, expects to post a fourth-quarter decline of as much as 9% due primarily to the weather. **Vail Resorts Inc.** says skier visits through Jan. 10 were down 7% due to the second-weakest snowfall through Christmas in Vail, Colo., in 40 years. **FirstEnergy Corp.**, an Akron, Ohio, utility, posted per-diluted-share net income of 41 cents for the fourth quarter, 15 cents to 20 cents below what earnings would have been under normal weather conditions, Chairman Willard R. Holland said. Many utilities had similar outcomes.

This is where providers of weather derivatives step in. "Our best marketing tool is companies' annual reports," Worldwide Weather Trading Managing Director Andrew Freeman says.

Wicor Inc.'s Wisconsin Gas Co. subsidiary used weather derivatives to salvage sales in the warm winter of 1997-98. **El Paso**

tion materials. A hot summer is bad for farmers and good for electric utilities. If weather derivatives traders can match up the winners and losers to structure swaps or sell options, the market potentially could be enormous.

"Weather is not just an environmental issue; it is a major economic factor. At least one trillion dollars of our economy is weather sensitive," Commerce Secretary William Daley told Congress last year when testifying on the National Weather Service's budget.

Weather derivatives differ from the policies insurance companies sell to cover catastrophes such as droughts, floods and hurricanes. Weather derivatives are used for more common occurrences, such as a cool summer that could knock 10 cents a share off an electric utility's profits or 10% off a clothier's sales. Derivatives don't pay off on losses, as insurance policies do; their payouts are determined by how much average temperatures, rainfall, snowfall, wind or other weather conditions vary in a specific location over a given time. And weather derivatives, unlike insurance policies, can later be sold to third parties.

Several insurance companies have started offering weather options and are writing some insurance policies that combine derivatives with some of the tax and accounting advantages of insurance. But most insurance companies won't do swaps. Swaps go against their business culture, which is based on paid premiums, Mr. Mercer says.

Energy companies account for the most use of weather derivatives. Electric and gas utilities are motivated by pressure

Weather derivatives differ from the policies insurance companies sell to cover catastrophes such as droughts, floods and hurricanes. Weather derivatives are used for more common occurrences, such as a cool summer that could knock 10 cents a share off an electric utility's profits or 10% off a clothier's sales.

Toro S'No Risk is Back!

Chicago Sun-Times TUESDAY, DECEMBER 1, 1998

It's Time For SNOW!

Prepare yourself for the winter season with a dependable TORO® Snowthrower.

TORO

S'no Risk

GUARANTEE*

If it doesn't snow, we'll return some dough!

Enter the **TORO** & **Chicago Sun-Times**

Guess the Snowfall Contest WIN

Details Inside!

- One of Two TORO® Snowthrowers!
- A Pair of Hockey Tickets!

Special Advertising Supplement



“If it doesn’t snow, we’ll return your dough!”

Toro Sno Risk Promotion Details

<http://www.toro.com/home/snowthrower/snoris/>



TORO S'no Risk

Qualifying Single Stage Models:

- 30170 CCR[™] Powerlite[®] R
- 30175 CCR[™] Powerlite[®] E
- 38400 CCR[™] 1000 R
- 38405 CCR[™] 1000 E
- 38409 CCR[™] 2400 GTS[®] R (R-Tek[™])
- 38410 CCR[™] 2400 R
- 38414 CCR[™] 2400 GTS[®] E (R-Tek[™])
- 38415 CCR[™] 2400 E
- 38420 CCR[™] 2500 R
- 38422 CCR[™] 2500 GTS[®] R (R-Tek[™])
- 38424 CCR[™] 2500 GTS[®] E (R-Tek[™])
- 38425 CCR[™] 2500 E
- 38430 CCR[™] 3000 GTS[®] R (Suzuki)
- 38435 CCR[™] 3000 GTS[®] E (Suzuki)

Qualifying Two-Stage Models:

- 38062 622 Power Throw[™]
- 38072 724 Power Throw[™]
- 38083 824 XL Power Throw[™]
- 38540 824 Power Shift[®]
- 38542 824 Power Shift[®] (OHV)
- 38556 1028 Power Shift[®]
- 38558 1028 Power Shift[®] (OHV)
- 38590 1232 Power Shift[®]

[Back to snowthrowers page](#)

Introducing the Toro S'no Risk[™] Promotion.

Purchase a new Toro Snowthrower between 7/31/98 and 01/31/99 from a participating dealer*, and receive up to 50% of the suggested promotional purchase price back if it doesn't snow**.

If it does snow you'll have a Toro snowthrower at promotional retail savings of up to \$200 off MSRP to help you clear your walks and driveways.

If It Doesn't Snow We'll Return Your Dough!**

- If your actual snowfall is less than 40% of the historical snowfall, receive a 20% refund.
- If your actual snowfall is less than 30% of the historical snowfall, receive a 30% refund.
- If your actual snowfall is less than 20% of the historical snowfall, receive a 40% refund.
- If your actual snowfall is less than 10% of the historical snowfall, receive a 50% refund.

*Not available everywhere. Subject to dealer participation.
**Snowfall based upon the dealers assigned reporting weather station.
See your participating dealer for details.

Old and New Refund Schedules

IF IT DOESN'T SNOW WE'LL RETURN YOUR DOUGH! AND YOU KEEP THE SNOWTHROWER. 	If it snows less than	You keep the Toro and you receive:
	20% * AVERAGE SNOWFALL	100% REFUND of suggested retail price
	30% * AVERAGE SNOWFALL	70% REFUND of suggested retail price
	40% * AVERAGE SNOWFALL	60% REFUND of suggested retail price
	50% * AVERAGE SNOWFALL	50% REFUND of suggested retail price

If between 7/1/98 and 4/30/99 it snows less than:	You keep the Toro and receive:
10% * Average Snowfall	50% REFUND OF suggested promotional price
20% * Average Snowfall	40% REFUND OF suggested promotional price
30% * Average Snowfall	30% REFUND OF suggested promotional price
40% * Average Snowfall	20% REFUND OF suggested promotional price

Regret in Decision Making

- A decision maker experiences *regret* if after having chosen an alternative and learned of the outcome, the decision maker wishes he or she had chosen another alternative.
- Notes
 - ❑ Regret is distinct from hindsight bias
 - ❑ What role does *anticipated regret* have in decision making?
 - ❑ How does regret explain the attractiveness of *price protection*?



Identify some situations in which you think avoiding regret is an important driver?

Regret in Action



- Dutch Postcode Lottery

- ❑ The Dutch postcode identifies 25 adjacent addresses (e.g., “1018 WB”)
- ❑ One postcode is drawn each week
- ❑ Everyone in the postcode who bought a ticket wins a (substantial) cash prize

- Ads

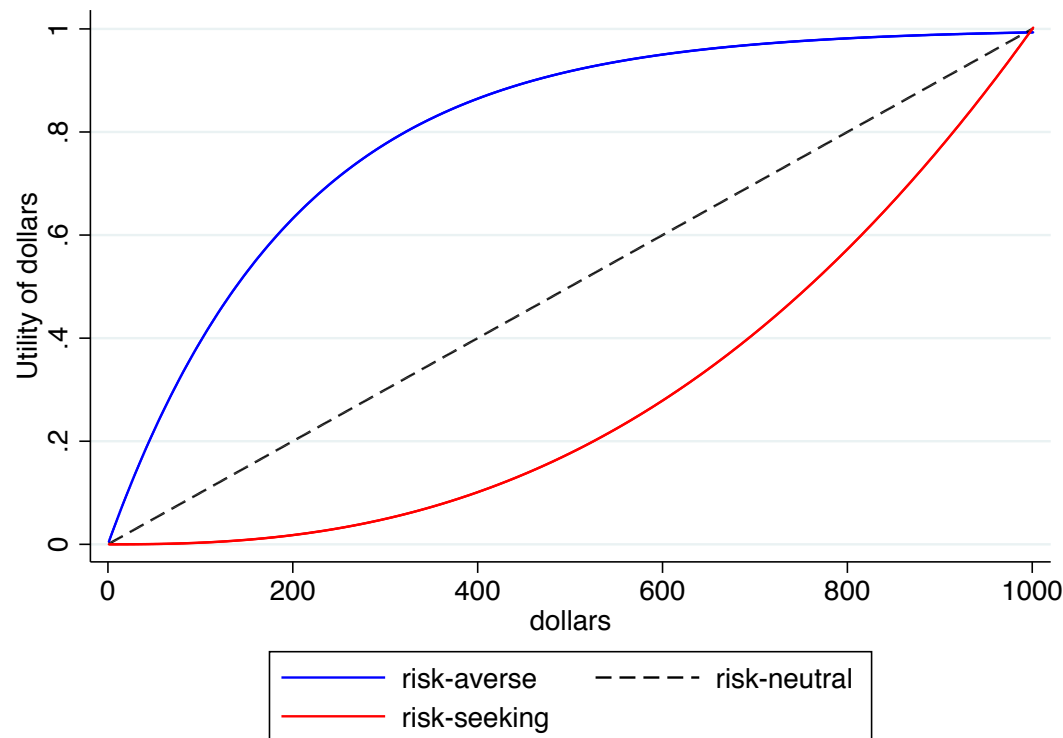
- ❑ “Don’t you have any tickets? Then your neighbors will win everything. So make sure that you buy some now.”
- ❑ “Sour, that is how it feels when you just miss an amount of at least 2 million. Having a multimillion prize fall on your own address, but winning nothing because you did not buy a ticket--that is something you do not want to experience.”

Expected Utility Theory

■ Normative Theory

□ Value by its *expected utility*

- The value of a risky prospect that offers outcome x_i with probability p_i is $p_1u(x_1)+\dots+p_nu(x_n)$



Prospect Theory

- Kahneman & Tversky (1979)
 - The value of a prospect, (p, x) , is given by $\omega(p)v(x)$ where
 - ω is a *decision weighting function* (indicates how are probabilities evaluated)
 - v is a *value function* (indicates how are outcomes evaluated)



Nine Decision Phenomena I

1. Endowment Effect (Kahneman, Knetsch, & Thaler, 1990)

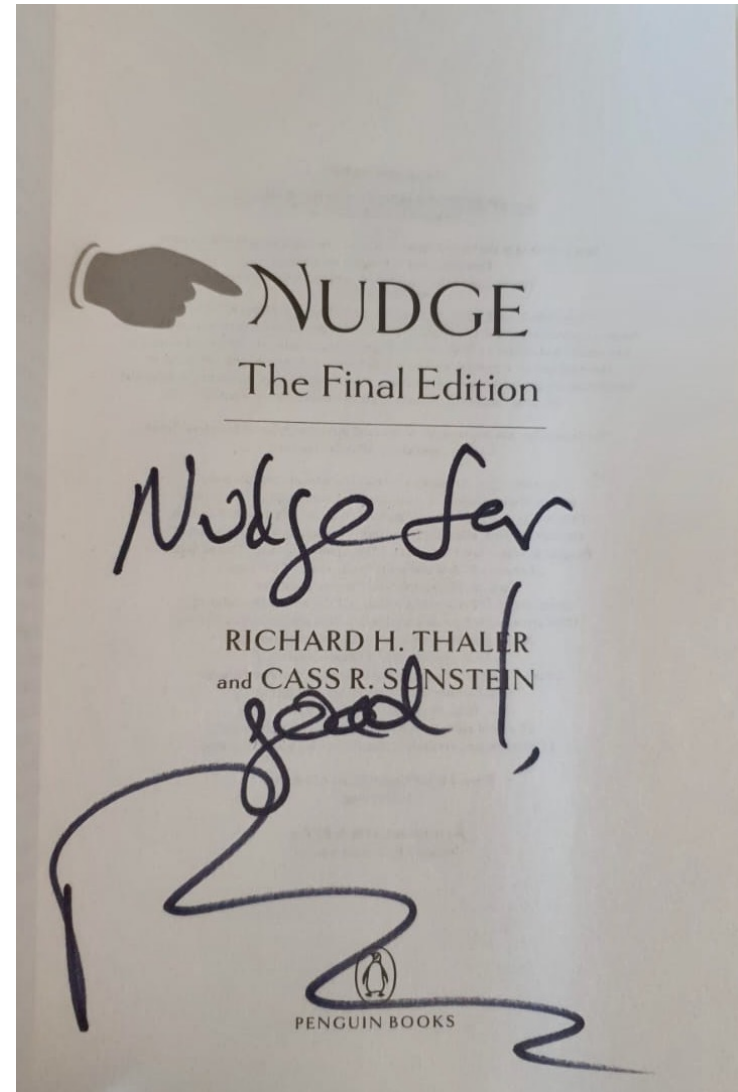
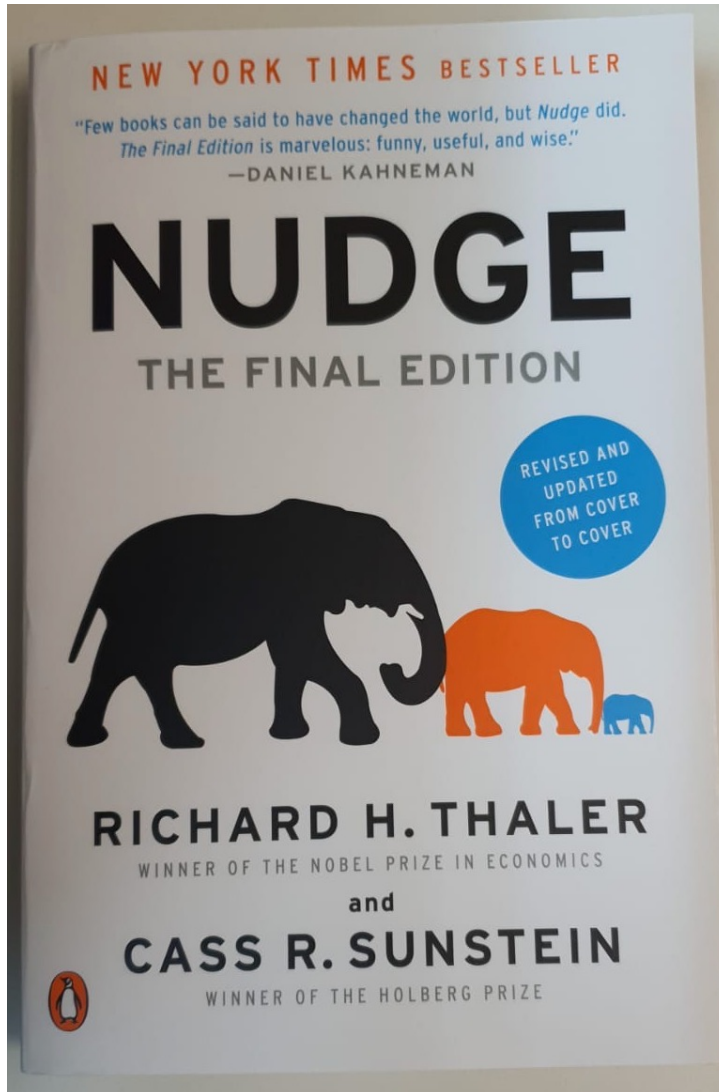
■ Setup

- 50% of students given mugs (sellers)
- 50% of students not given mugs (buyers)

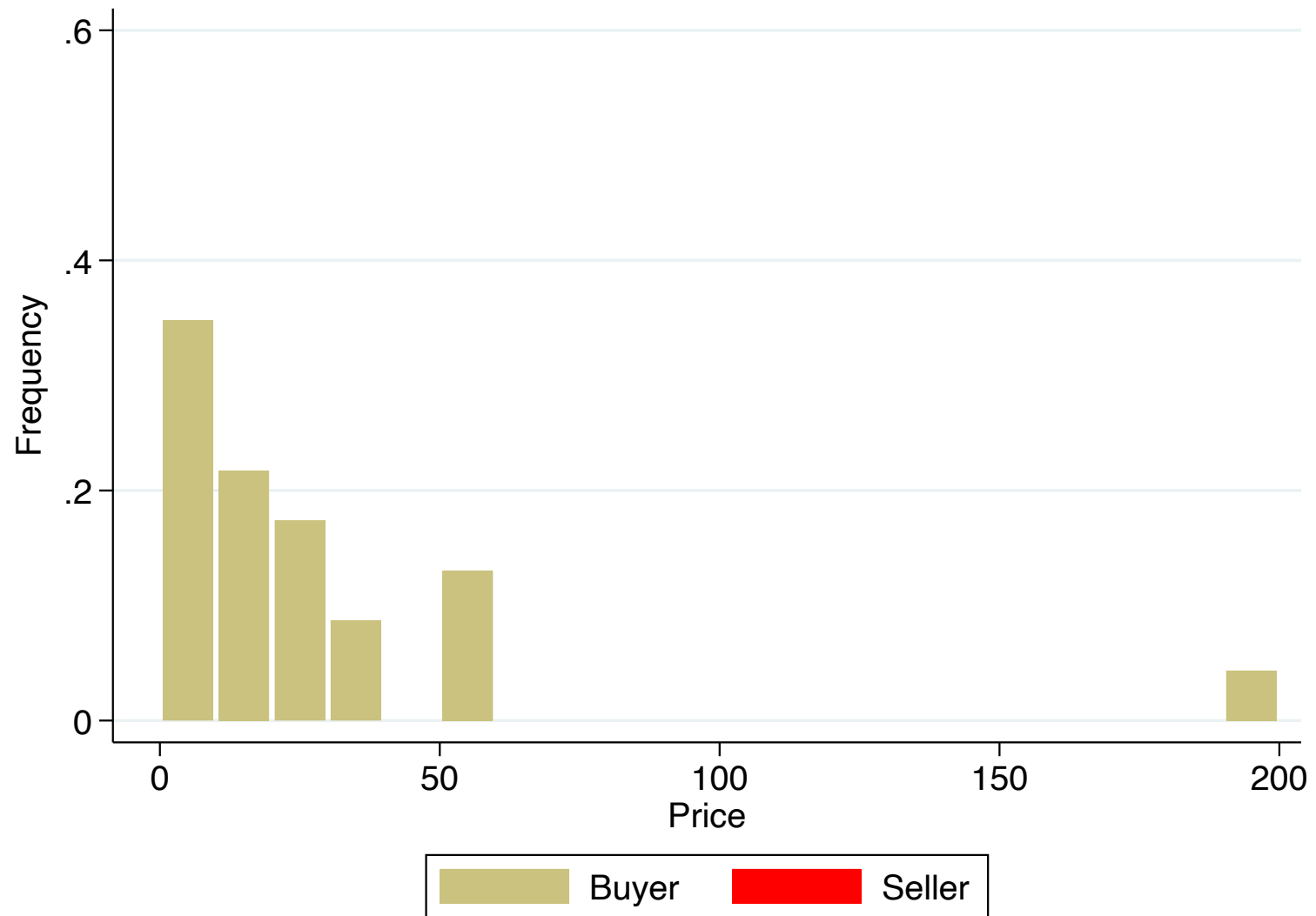
■ Results



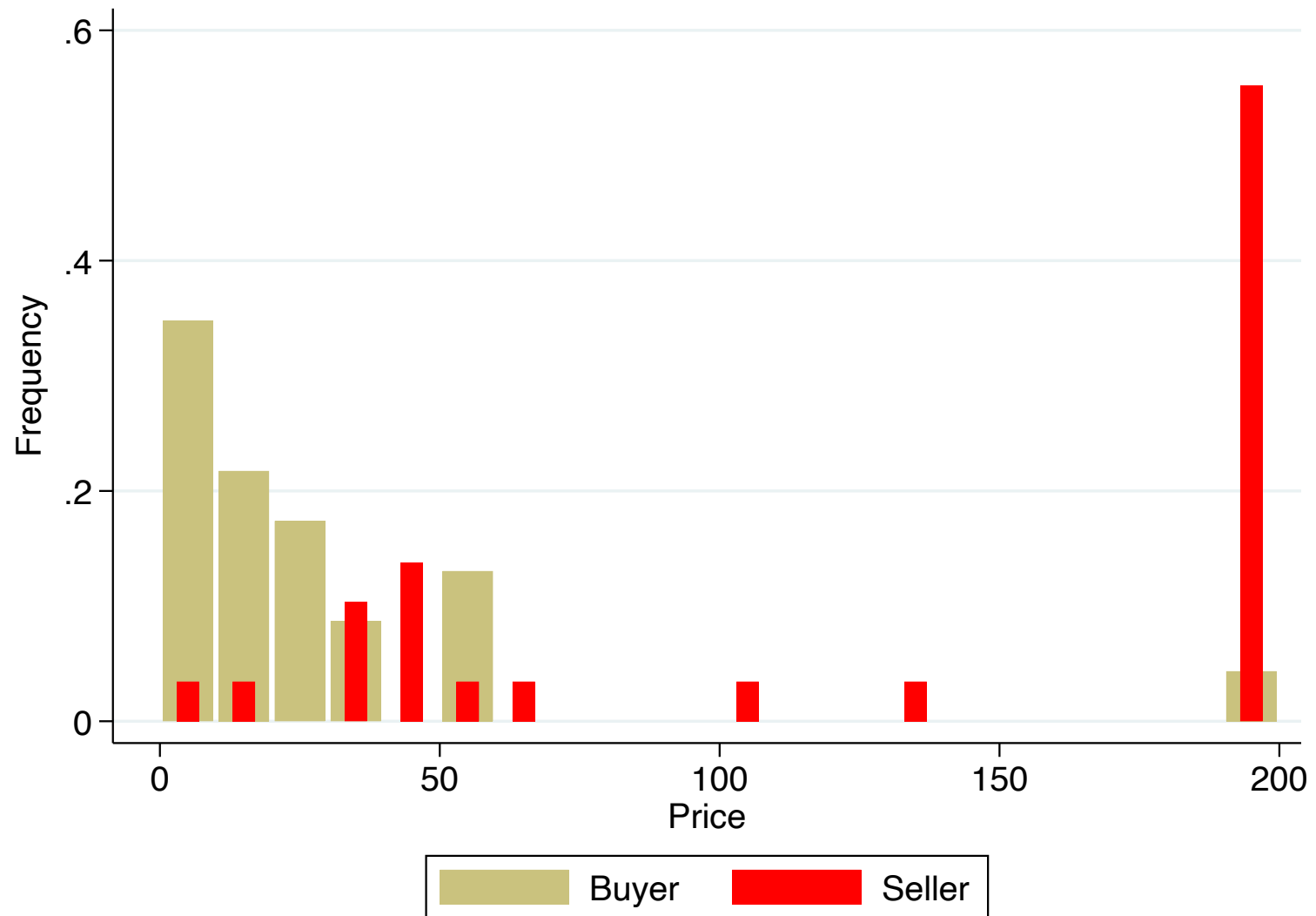
Time to make a decision!



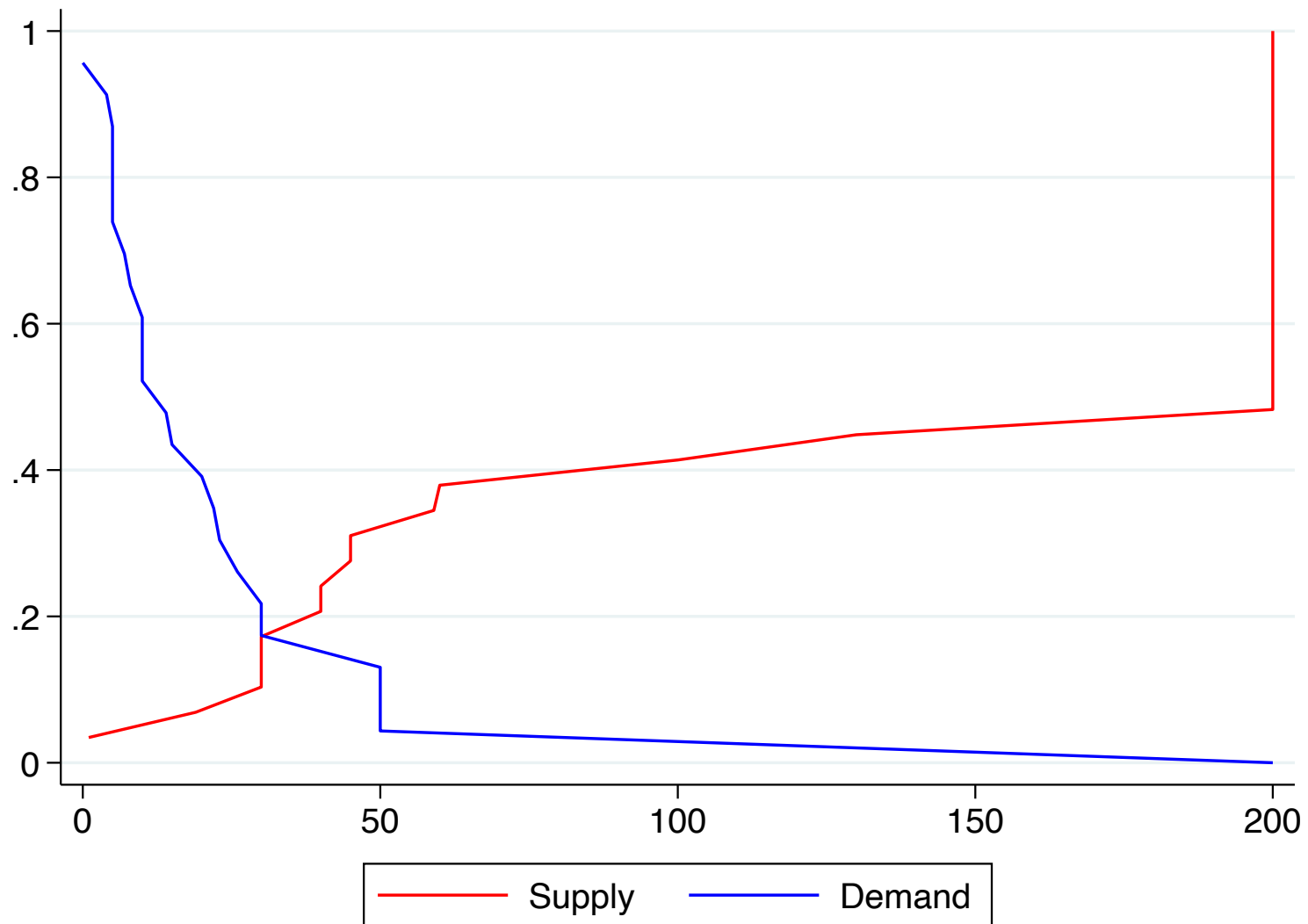
Prices!



Prices!



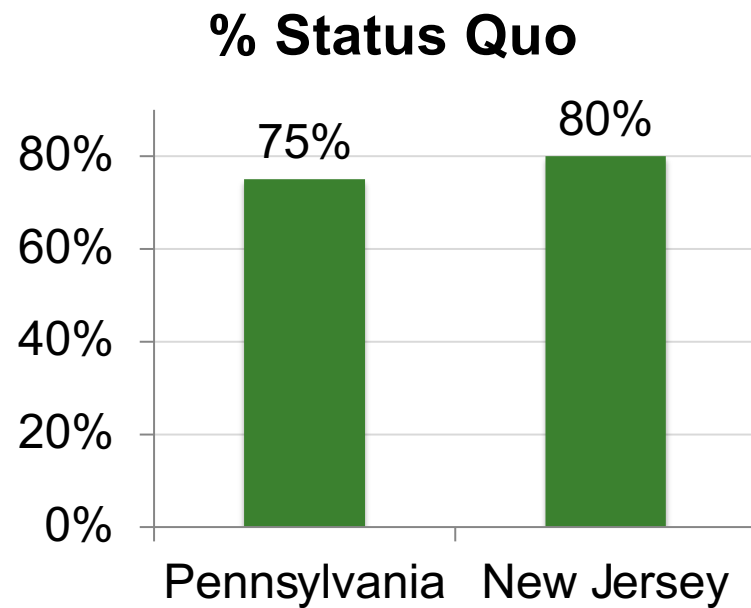
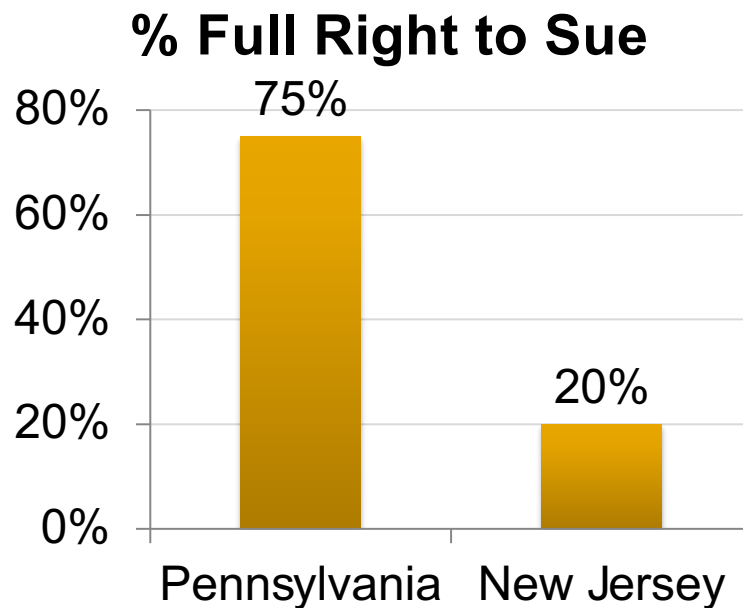
Supply and Demand (Axis Flipped)



Nine Decision Phenomena II

2. Status Quo (Johnson et al., 1993)

- Setup: In early 1990s, Pennsylvania and New Jersey introduced right to sue (at a cost)
 - Pennsylvania default: full right to sue (more options, expensive)
 - New Jersey default: restricted right to sue (fewer options, cheap)
- Results



Nine Decision Phenomena III

3. Preference for Increasing Income Streams

(Loewenstein and Sicherman, 1989)

- ❑ Setup: Subjects chose between hypothetical jobs that differed only in wage profile (total undiscounted wages fixed; decreasing, flat, increasing)
- ❑ Result 1:
 - 12% preferred “decreasing”
 - 12% preferred “flat”
 - 76% preferred “increasing”
- ❑ Result 2:
 - After being told that decreasing + savings dominated increasing, 69% preferred increasing

Nine Decision Phenomena IV

4. Goal Setting (Locke & Latham, 1979)

- Robust Result: higher goals lead to better performance



Nine Decision Phenomena V

5. Framing of Outcomes (Kahneman & Tversky, 1979)

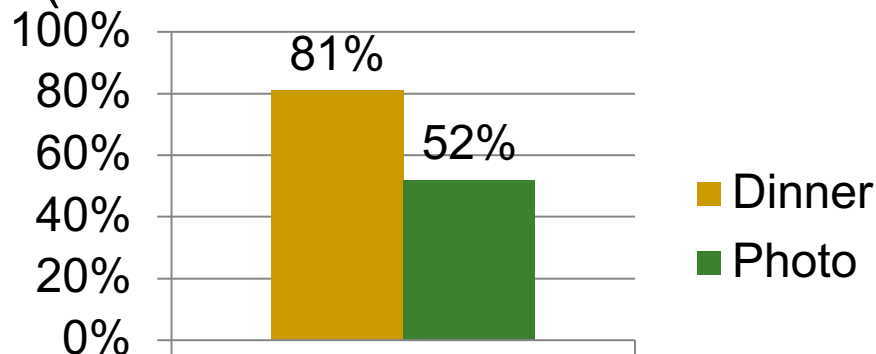
- Gain Frame

- In addition to whatever you own, you have been given \$1000. You have to choose between:
 - A sure gain of \$500
 - 50% chance to gain \$1000 and 50% chance to gain nothing

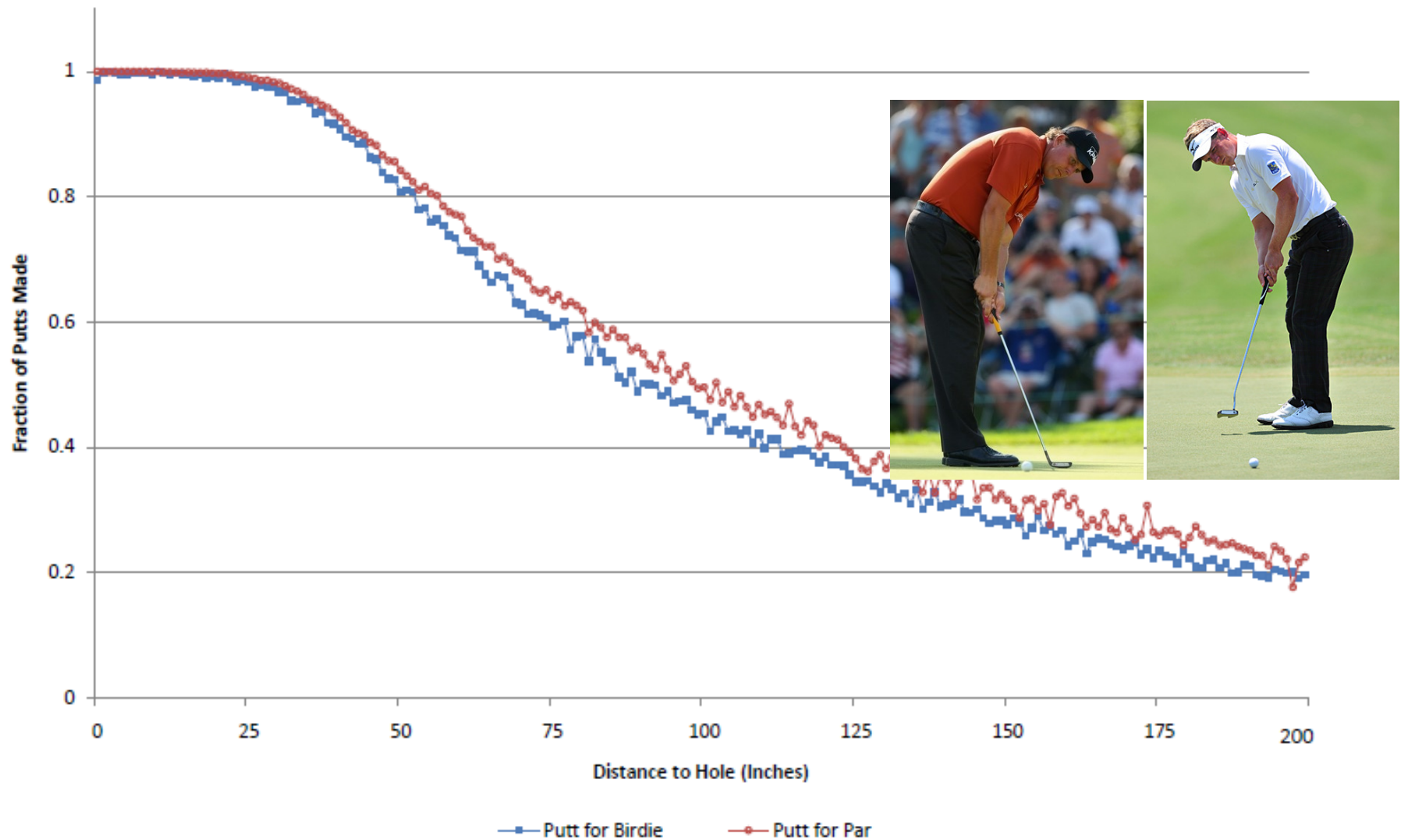
Nine Decision Phenomena VI

6. Improvements vs. Tradeoffs (Tversky & Kahneman, 1991)

- ❑ Subjects are given one of two prizes:
 - 1 free **dinner** at a nice restaurant + 1 Stanford calendar
 - 1-8 x 10 professional **photos** + 1 Stanford calendar
- ❑ Subjects can choose to exchange their original prizes for:
 - 2 free dinners at a nice restaurant
 - 1-8 x 10, 2- 5 x 7, and 3 wallet size professional photos
- ❑ Results (% who chose restaurant exchange)



Nine Decision Phenomena VII



Nine Decision Phenomena VIII

8. Marathon Finishing Times (Allen, Dechow, Pope, & Wu, 2017)

- ❑ Study
 - Finishing times of 9,789,083 marathoners
- ❑ Necessary fact
 - Runners tend to choose “round numbers” as goals
 - ❑ 28% select hour goals (3:00, 4:00, 5:00, etc.)
 - ❑ 49% select half hour goals (3:00, 3:30, 4:00, 4:30, etc.)
 - ❑ 67% select 10 minute goals (3:30, 3:40, 3:50, 4:00, etc.)

Last Sunday...

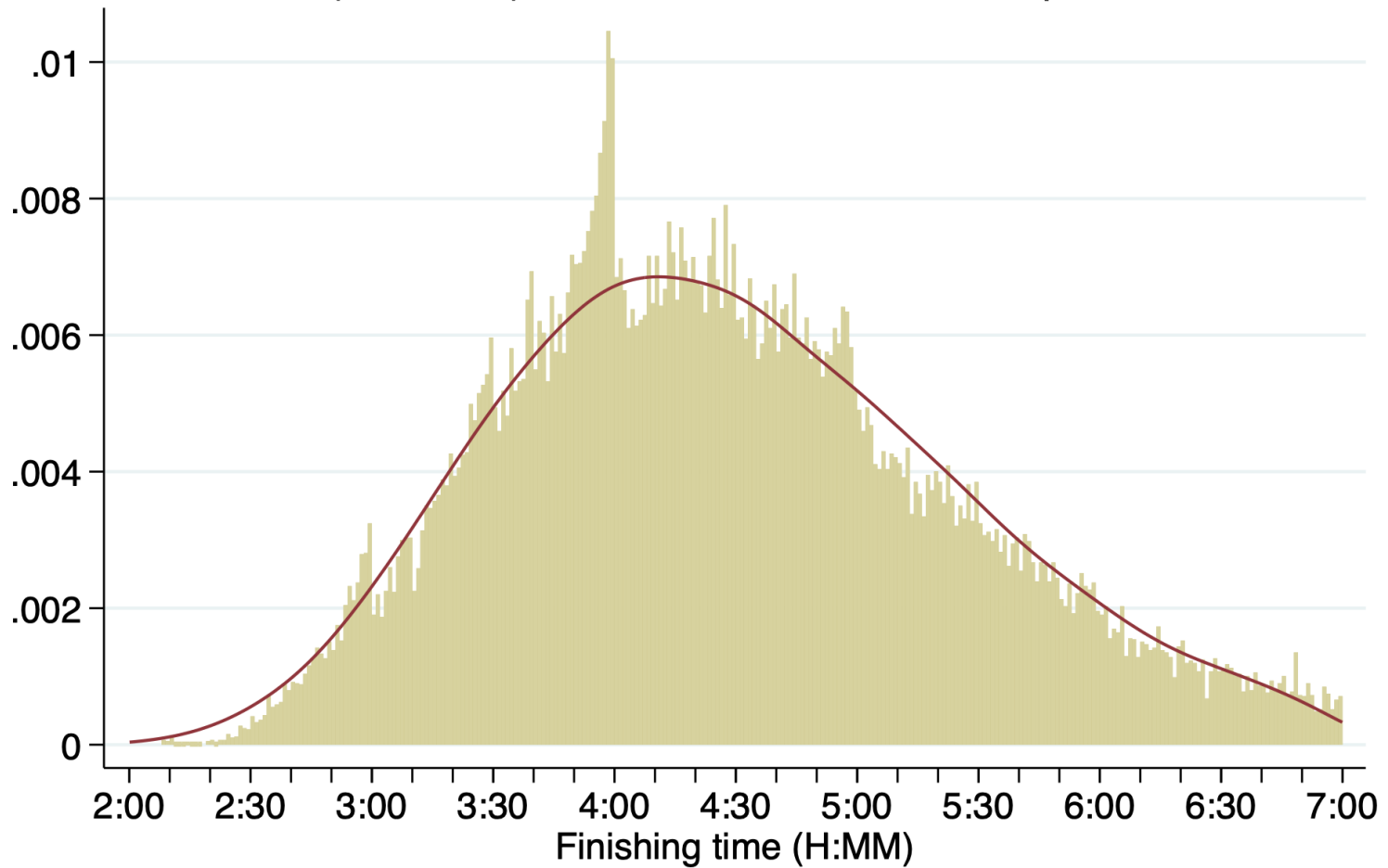


Last Sunday...



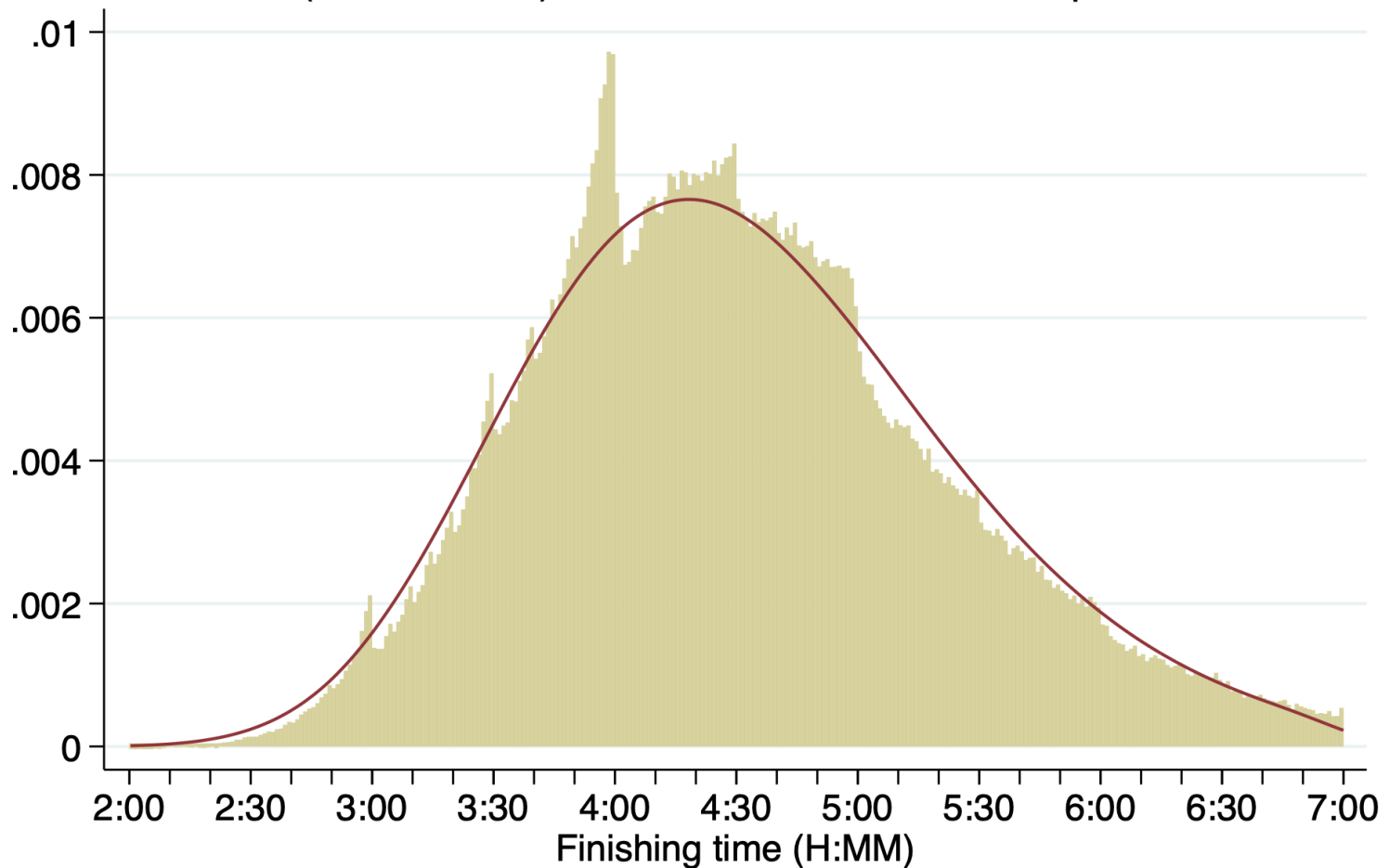
Last Sunday...

New York Marathon, 2025, Finishing Times
(n=59,131), Width of bins=1 minute, Chiptime



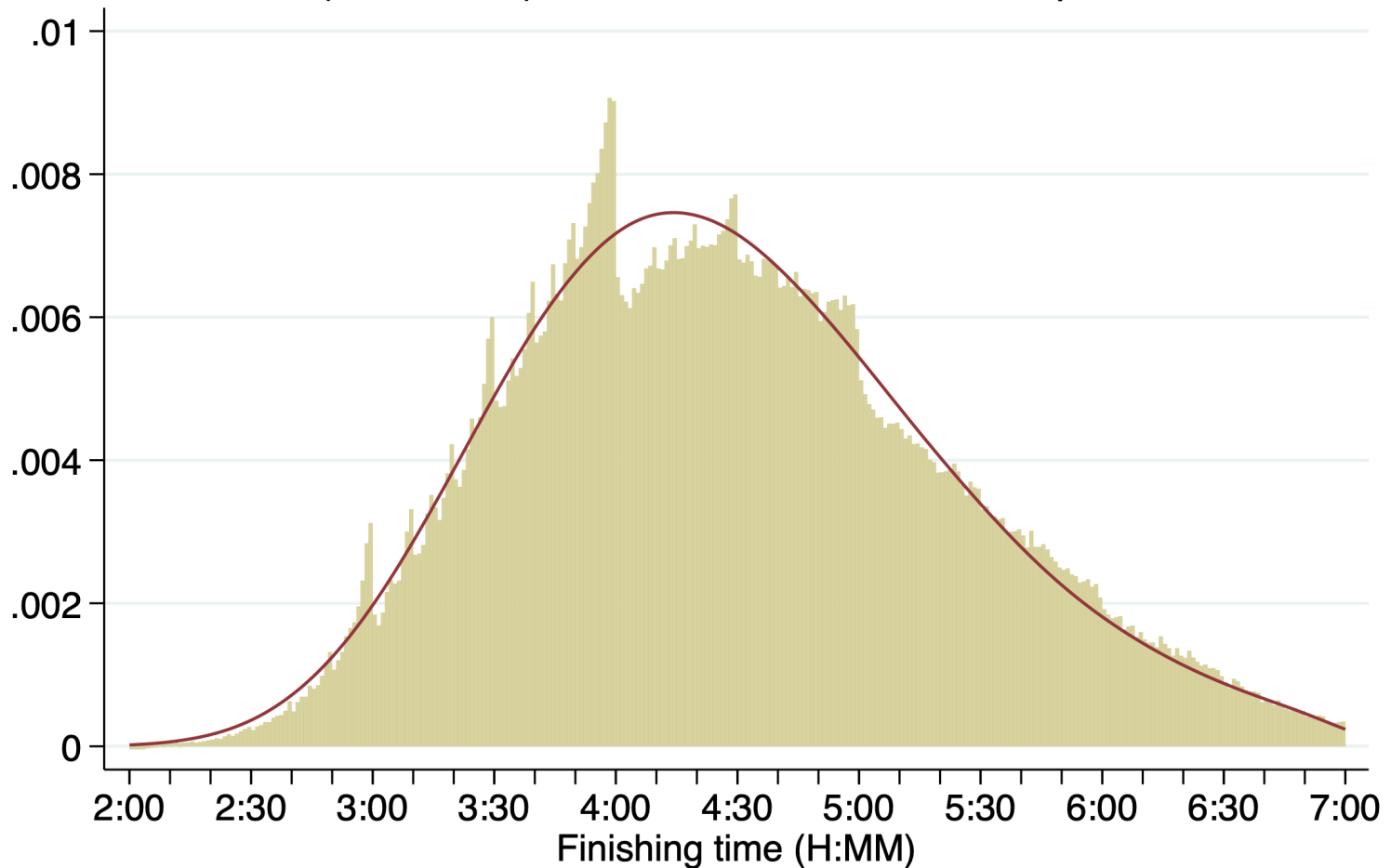
Many Sundays in New York...

New York Marathon, 1999-2025, Finishing Times
(n=1,012,350), Width of bins=1 minute, Chiptime



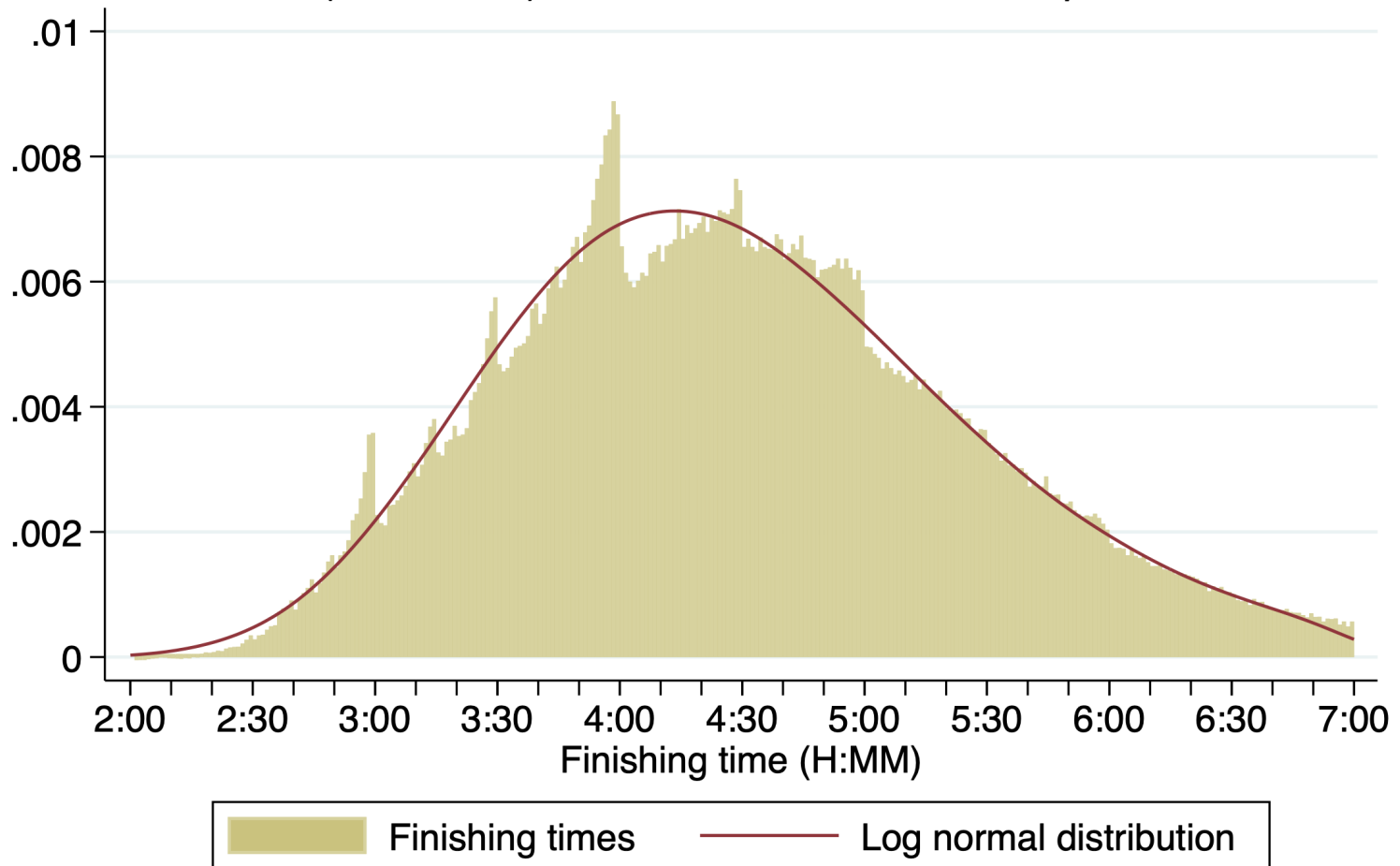
Many Sundays in Chicago...

Chicago Marathon, 1998-2025, Finishing Times
(n=972,270), Width of bins=1 minute, Chiptime



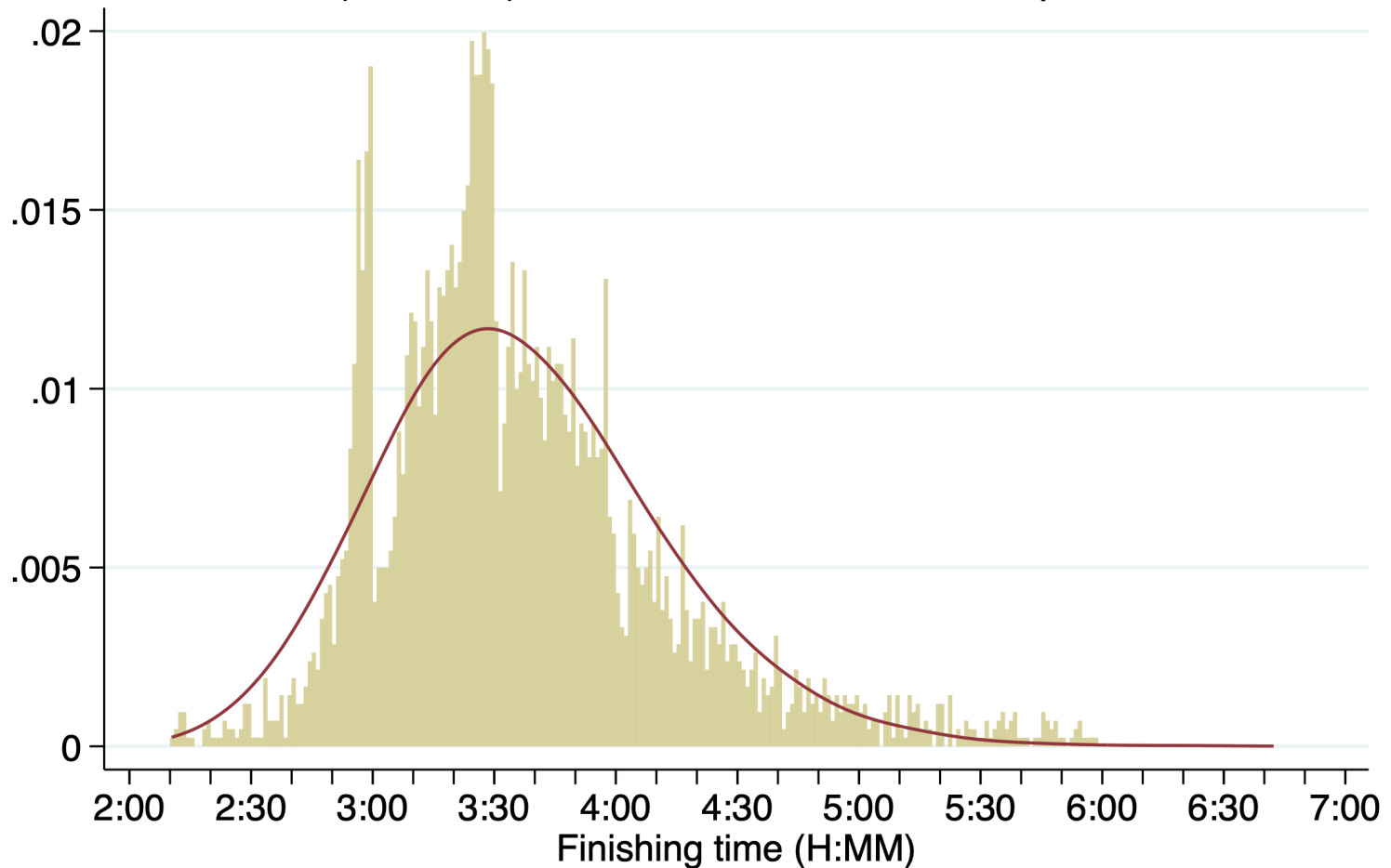
Many Sundays in London...

London Marathon, 2001-2025, Finishing Times
(n=913,159), Width of bins=1 minute, Chiptime



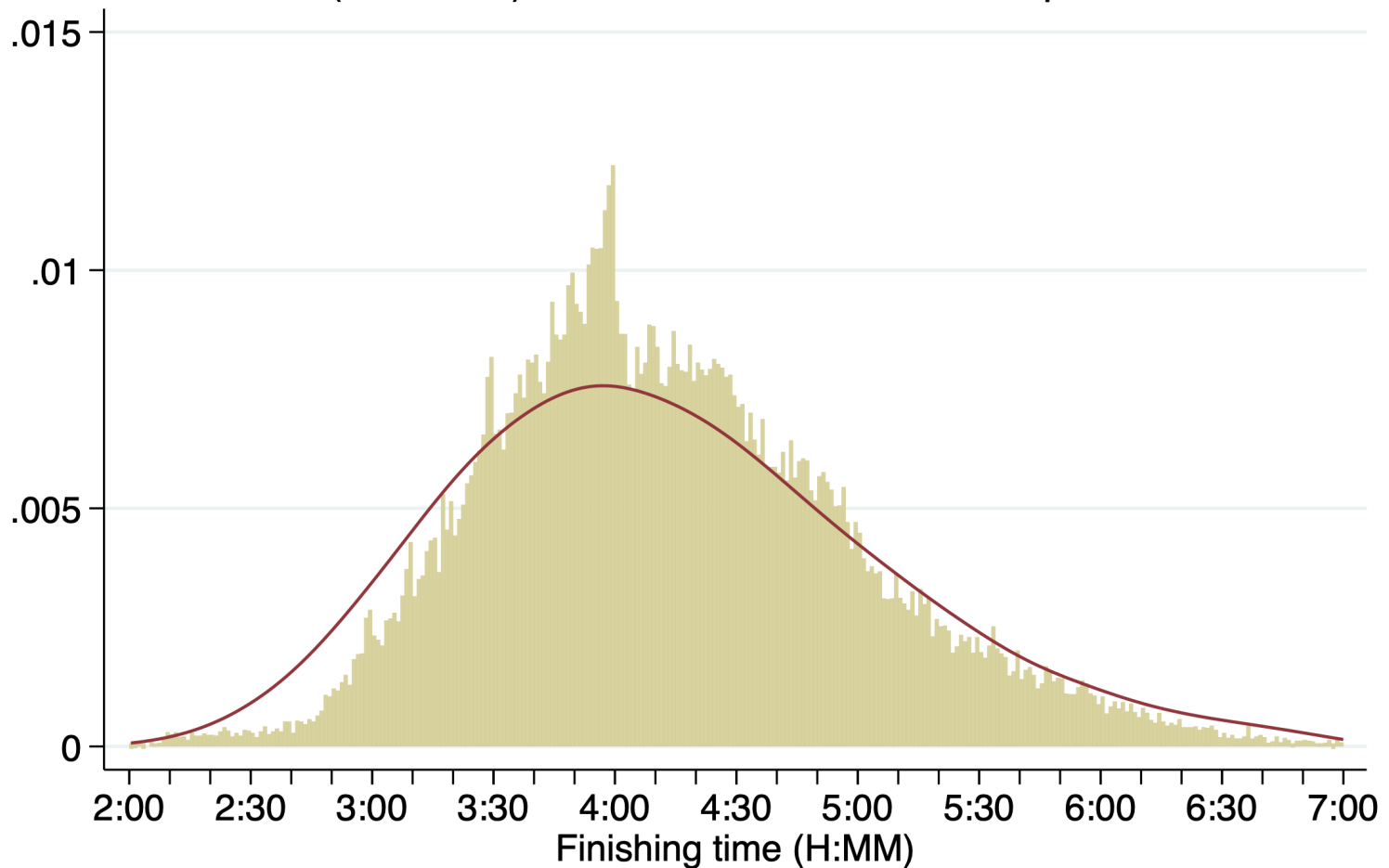
Many Sundays in Hong Kong...

Hong Kong Marathon, 2019-2024, Finishing Times
(n= 4,208), Width of bins=1 minute, Chiptime



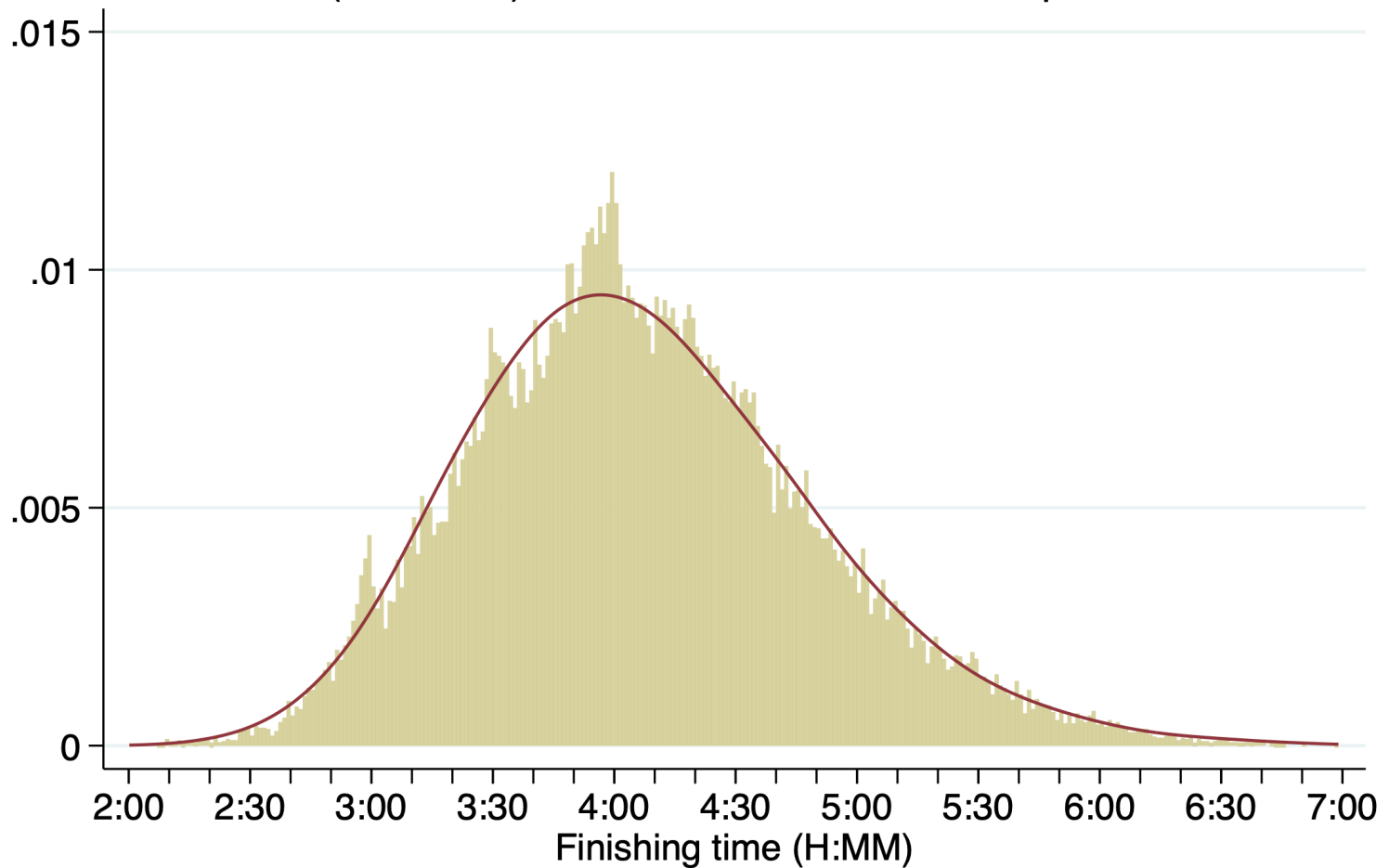
Many Sundays in Toronto...

Toronto Marathon, 2000-2025, Finishing Times
(n=66,998), Width of bins=1 minute, Chiptime

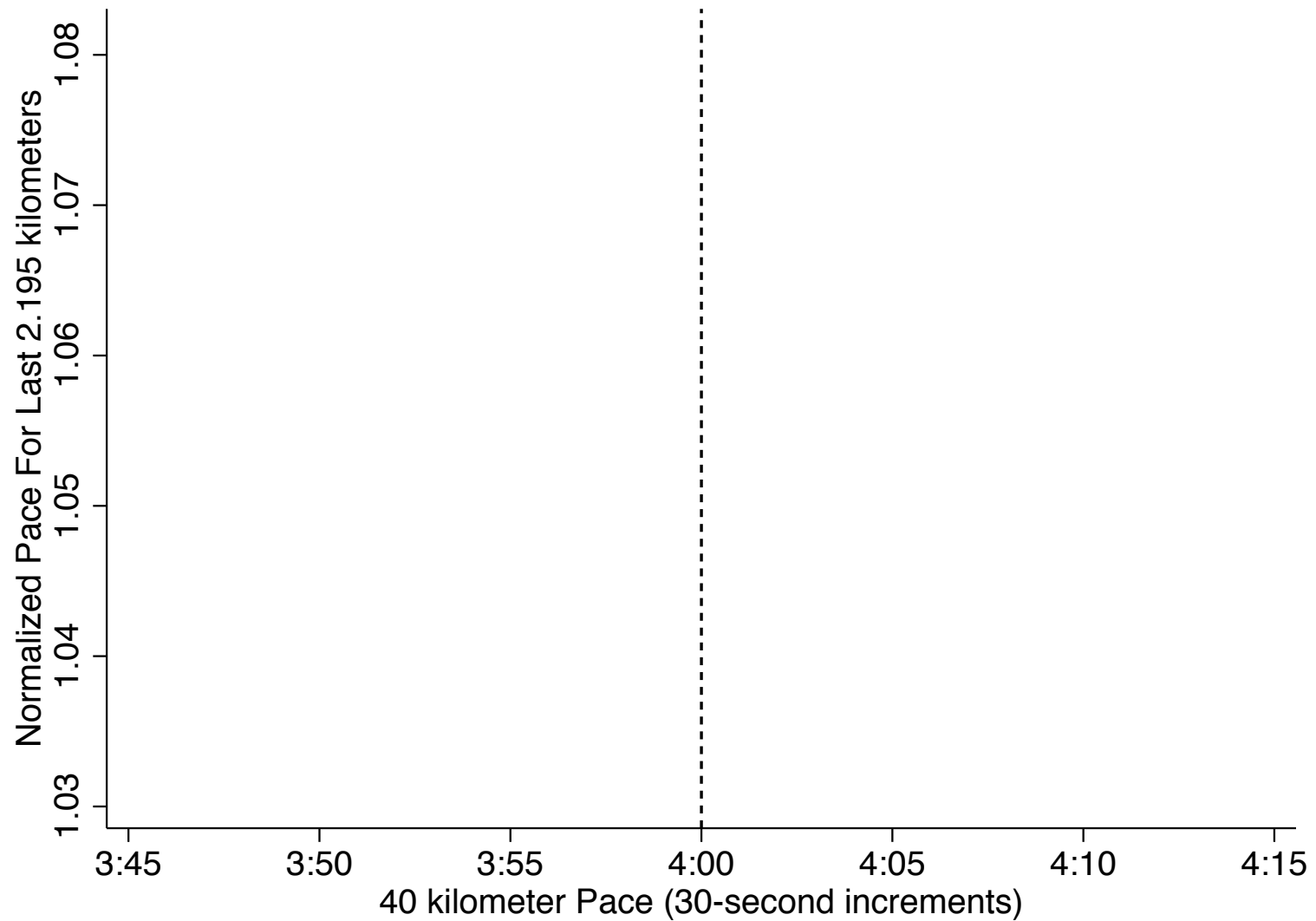


Many Sundays in Argentina...

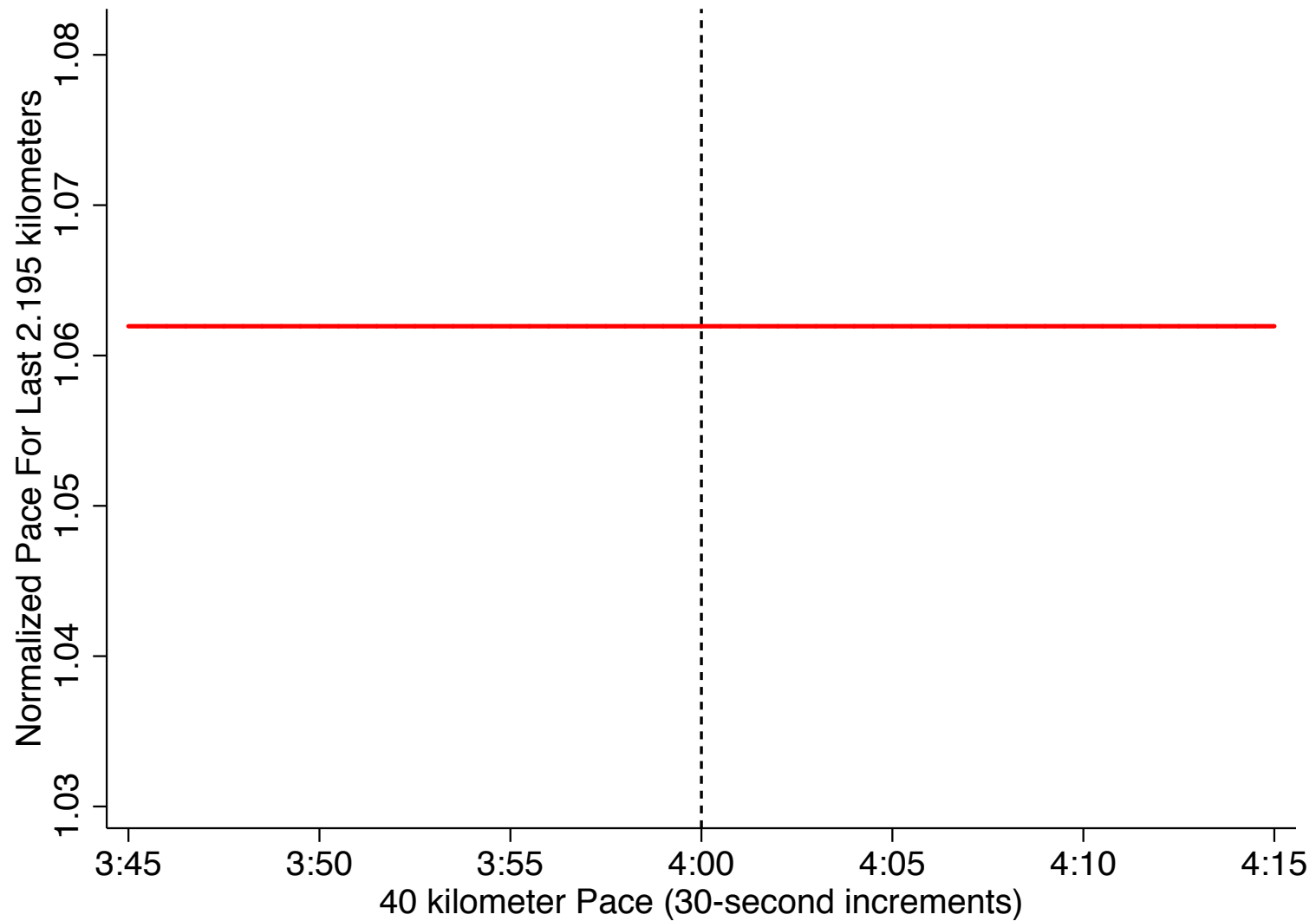
Buenos Aires Marathon, 2022-2025, Finishing Times
(n=42,726), Width of bins=1 minute, Chiptime



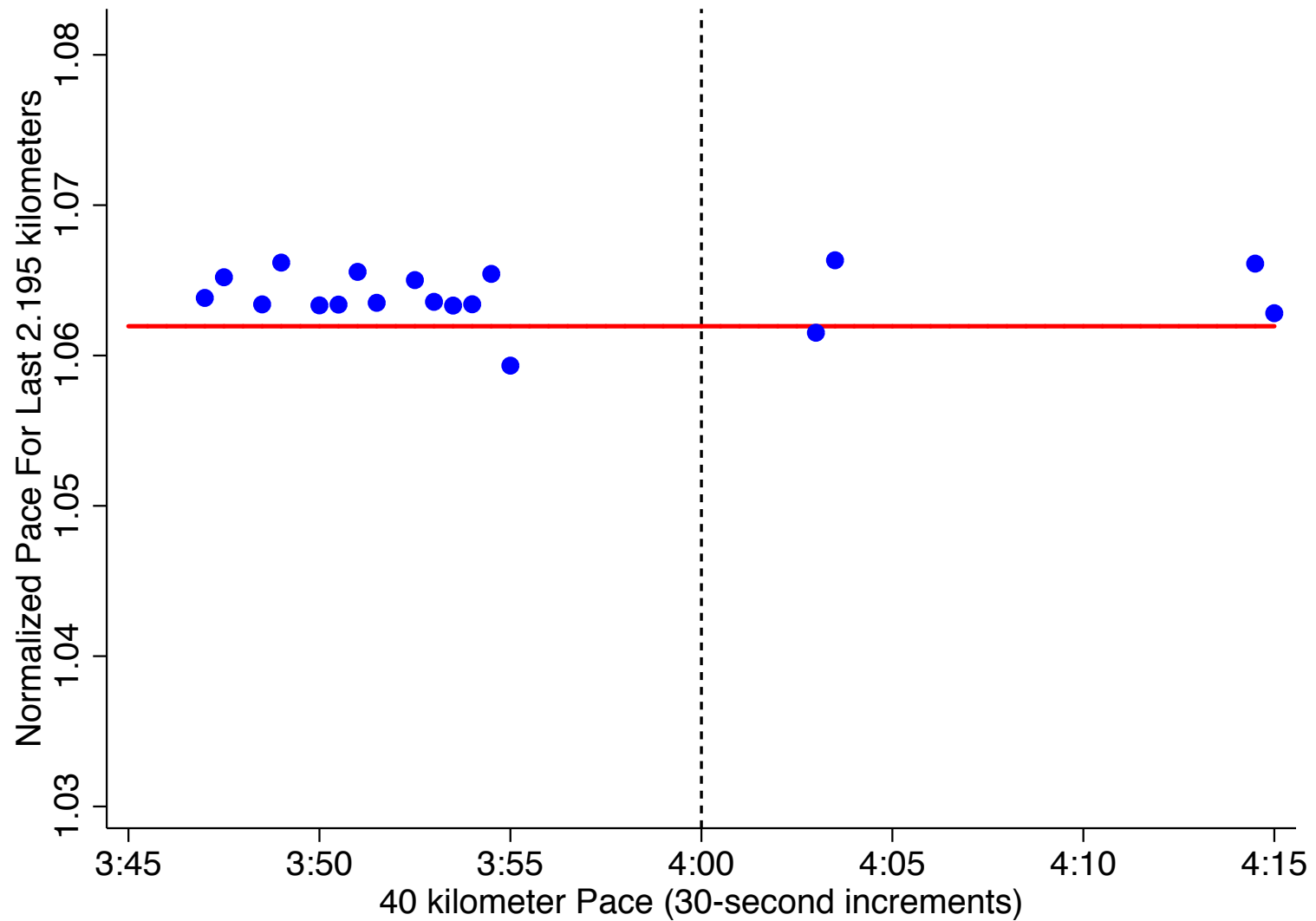
Effort at end of race



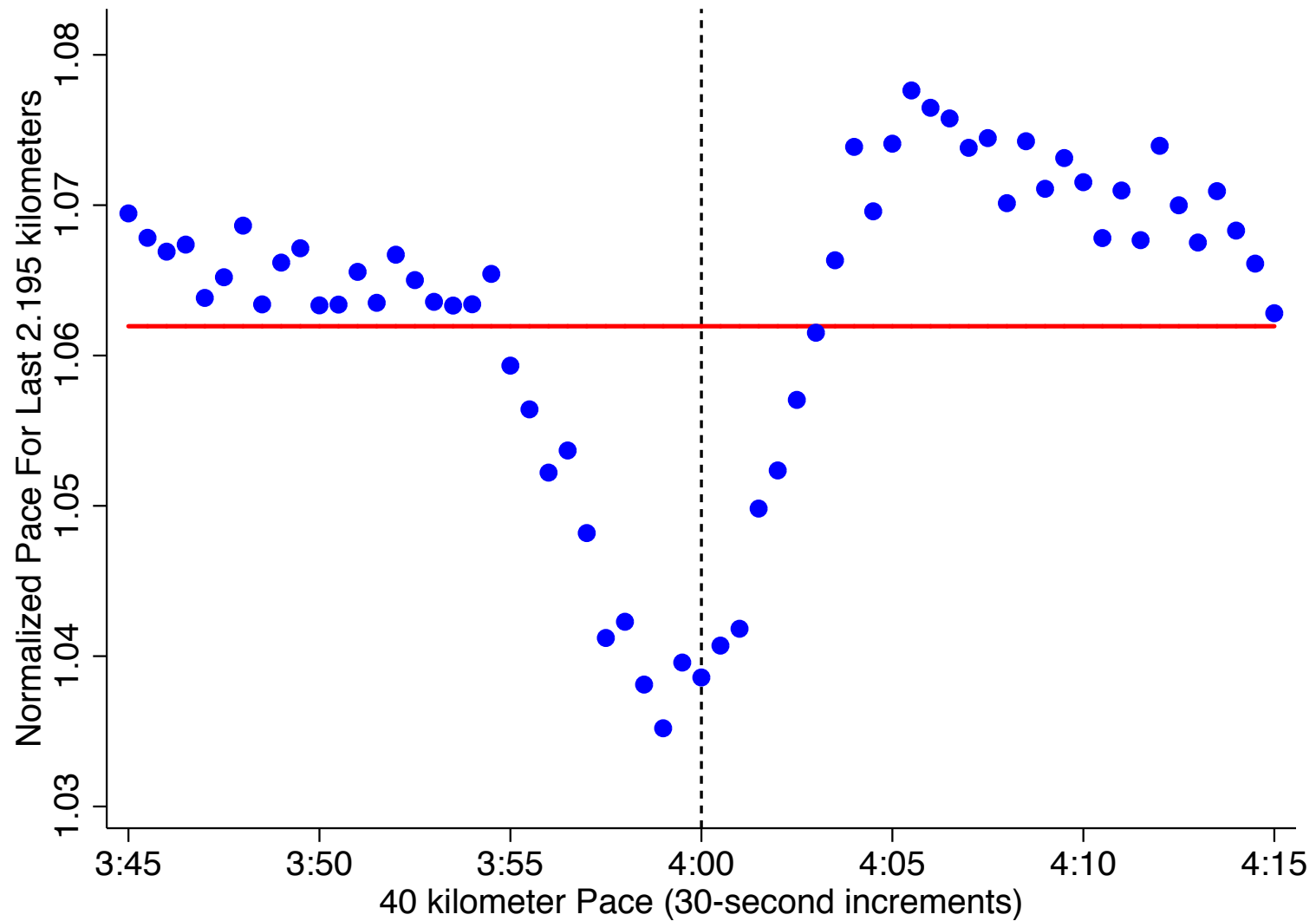
Effort at end of race



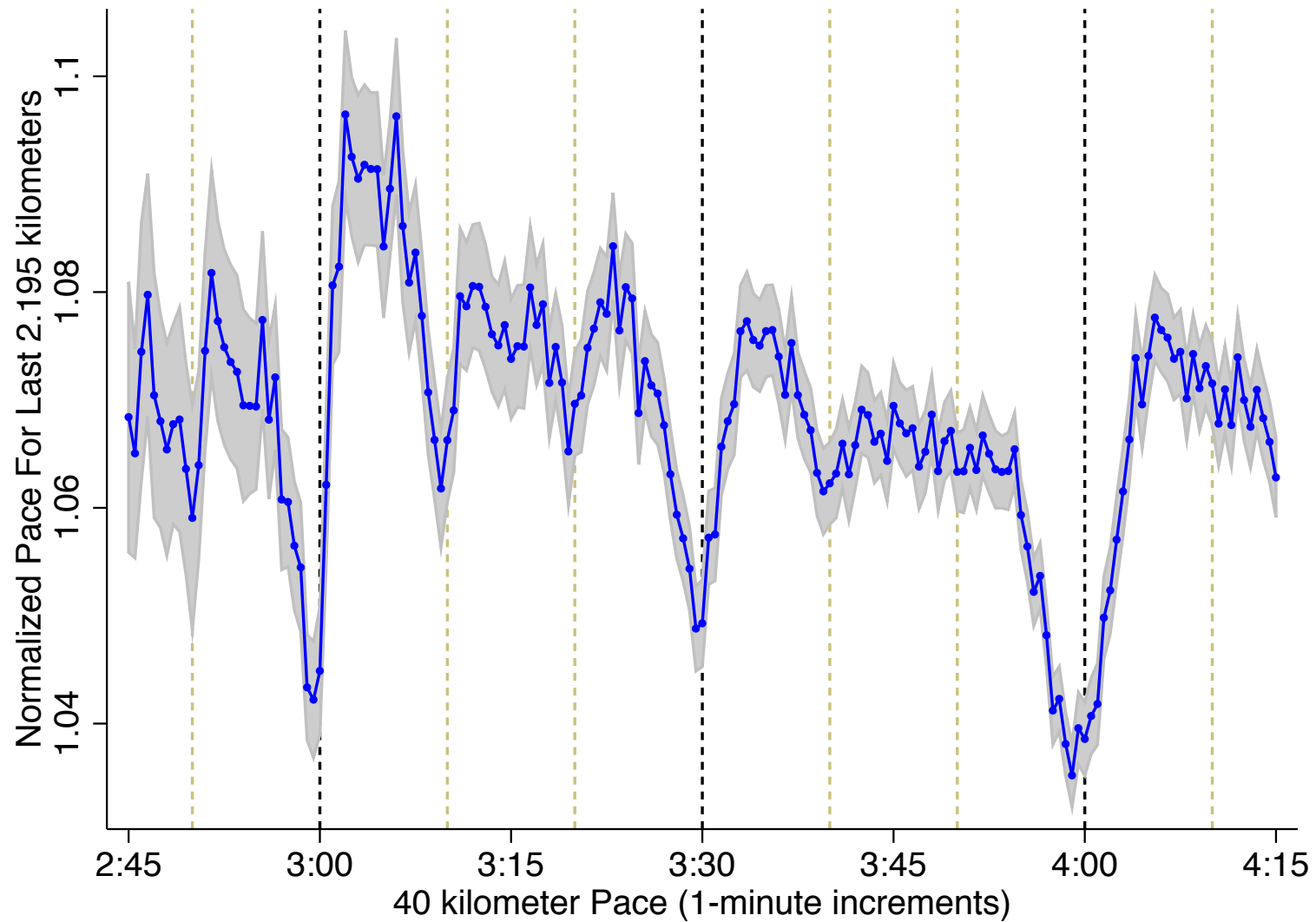
Effort at end of race



Effort at end of race



Effort at end of race



Nine Decision Phenomena IX

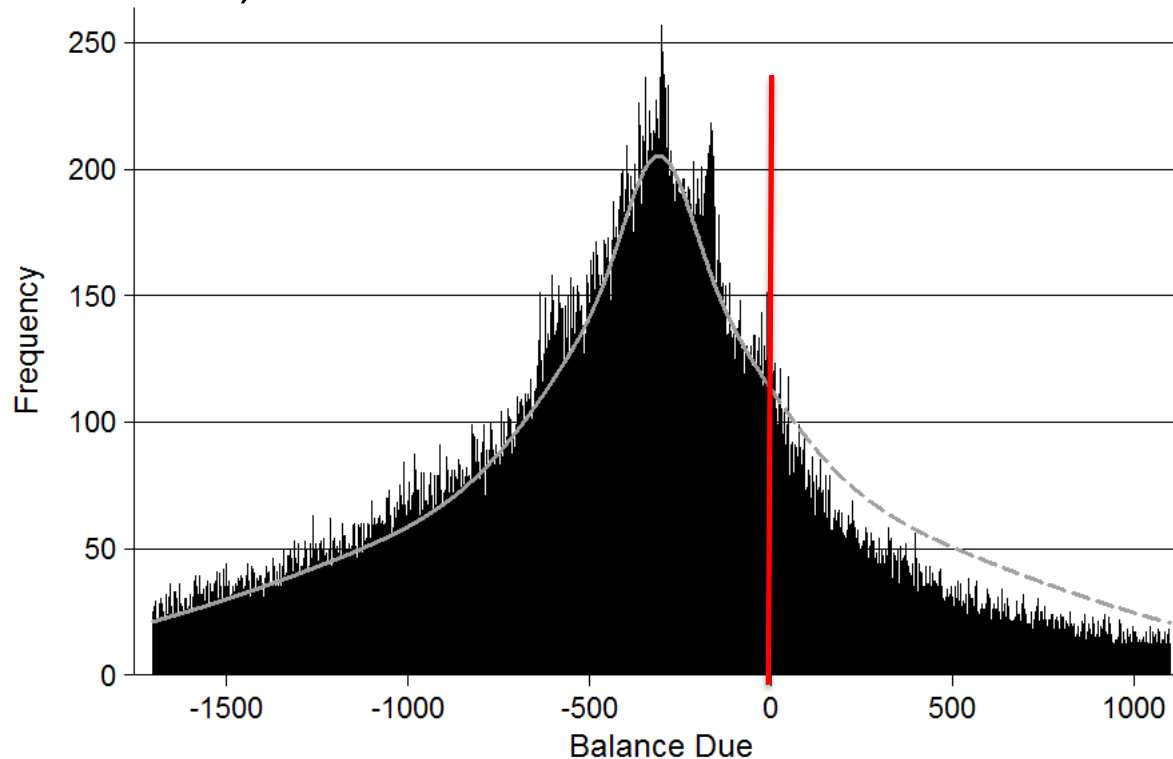
9. Tax “Sheltering” (Rees-Jones, 2017)

- ❑ Study
 - 1979-1990 US tax filers
 - 229,116 tax returns with non-zero tax liabilities
- ❑ Balance due upon filing
 - Withheld taxes plus
 - Adjustments
 - ❑ Other taxable income
 - ❑ Itemizing deductions
 - ❑ Business expenses
 - ❑ Tax-preferred savings vehicles
 - ❑ Etc.

Tax “Sheltering”

■ Finding

- 136% “excessive mass” around \$0 balance
 - Pattern holds for all quartiles of incomes (strongest for wealthiest)



Nine Decision Phenomena

- 1) Endowment Effect: Selling prices $\approx 2 \times$ Buying prices
- 2) Status Quo Bias: The status quo is attractive
- 3) Preference for Increasing Income Sequences
- 4) Goal Setting: Goals work
- 5) Framing of Outcomes: Risk seeking for losses, Risk aversion for gains
- 6) Improvements vs. Tradeoffs: Improvements are more attractive than Tradeoffs
- 7) Putting behavior: PGA golfers are much more likely to make par than birdie putts
- 8) Marathon finishing time: Finishing times “bunch” just before round numbers
- 9) Tax Sheltering: Tax balances “bunch” at \$0

Prospect Theory Value Function

■ Properties

- Outcomes are defined with respect to reference point (often status quo)

- “Losses loom larger than gains”:

$$|v(-x)| > v(x)$$

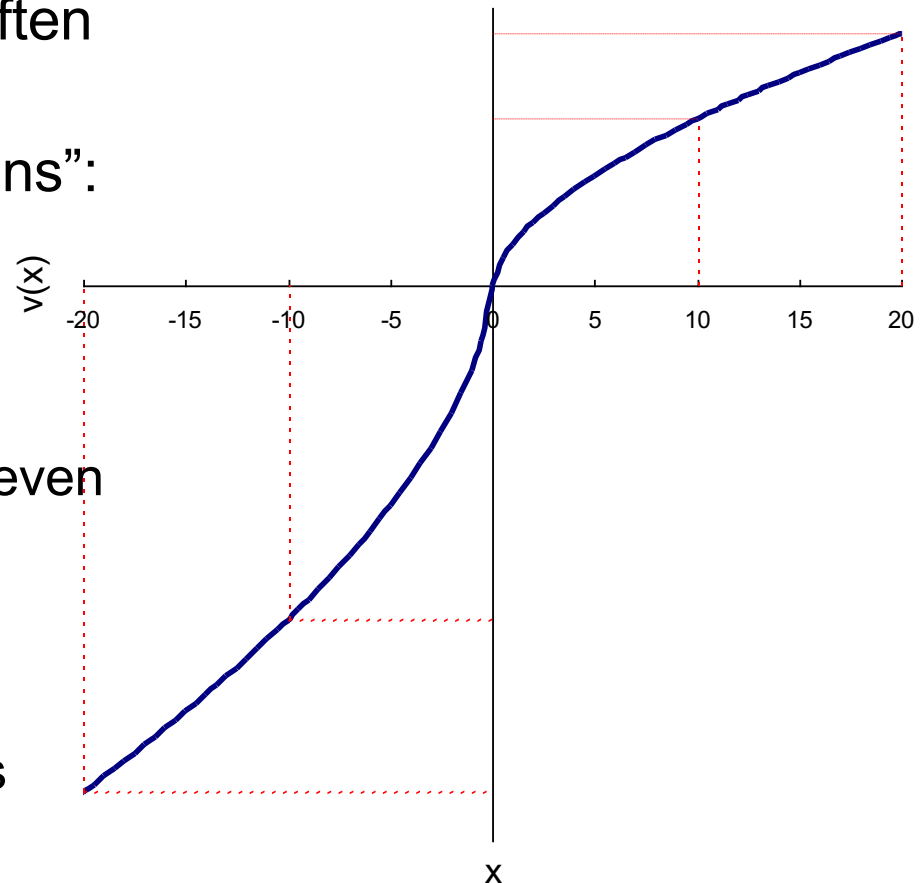
- Quasi-universal Constant:

$$|v(-x)| \approx 2v(x)$$

- Organizations are probably even more loss averse

- Diminishing sensitivity away from reference point

- function is concave for gains and convex for losses



Loss aversion and Bias for Status Quo

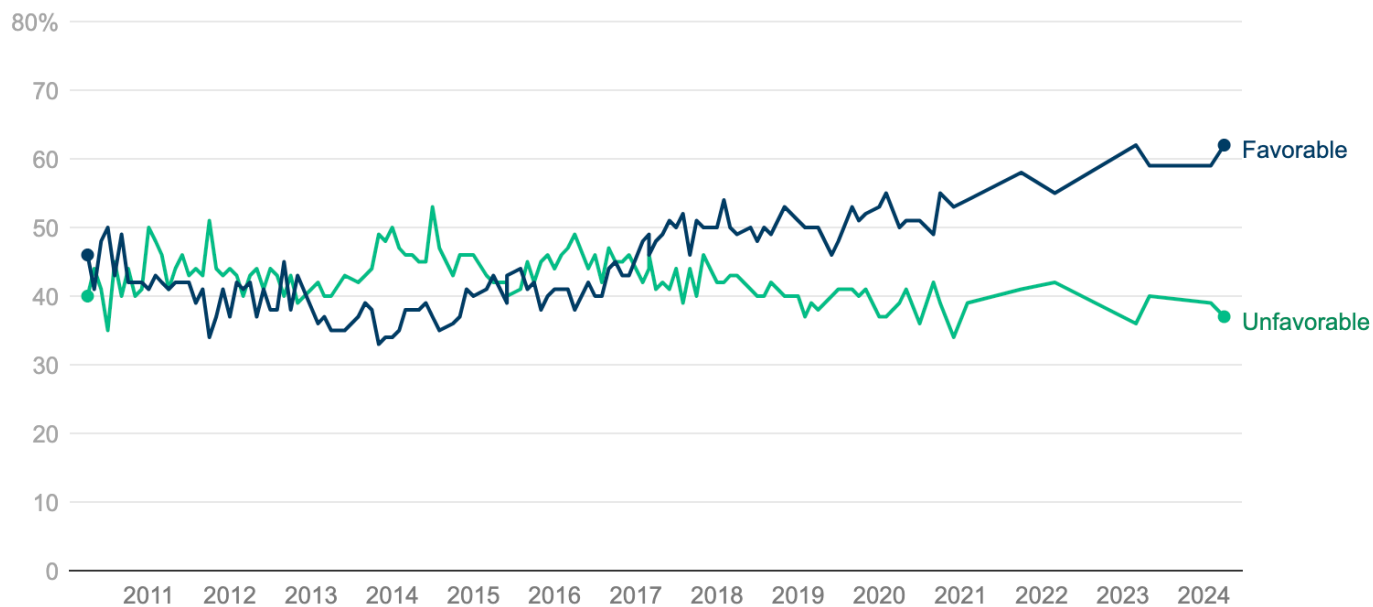
- Consider an organizational change
 - New office
 - Benefits plan
 - Organizational structure
 - Decision making process
- Most changes involve “upgrades” and “downgrades”
- What does prospect theory say about change?

An example: “Obamacare”

Figure 1

Clear Majority Of Public View The ACA Favorably

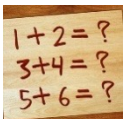
In general, do you have a favorable or unfavorable opinion of the health reform bill signed into law in 2010, known commonly as the Affordable Care Act or Obamacare?



NOTE: See topline for full question wording.

SOURCE: [KFF Health Tracking Polls](#) • PNG

KFF

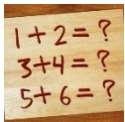


How might prospect theory explain, in part, the increase in popularity of ACA?

Off-line processing



Identify a situation in which you or your organization suffered from “status quo bias”?



How does prospect theory explain the 9 phenomena?



Identify some examples of situations in which people are risk-averse because losses loom larger than gains

Risky Choice Observations I

- Four-fold pattern of risk behavior (Tversky & Kahneman, 1992)

Probability	Gain		Loss	
	Gamble	Cash equivalent	Gamble	Cash equivalent
Low	.05 chance at \$100	\$14	.05 chance at \$-100	(\$8)
High	.95 chance at \$100	\$78	.95 chance at \$-100	(\$84)

risk-seeking
(small prob gains)

risk aversion
(small prob losses)

risk aversion
(most gains)

risk-seeking
(most losses)

Risky Choice Observations II

- Ambiguity aversion (Fox & Tversky, 1995)
 - Sum of price of *familiar* gambles
 - \$100 if high temperature in San Francisco > 60 degrees
 - \$100 if high temperature in San Francisco < 60 degrees
 - Sum of price *unfamiliar* gambles
 - \$100 if high temperature in Istanbul > 60 degrees
 - \$100 if high temperature in Istanbul < 60 degrees

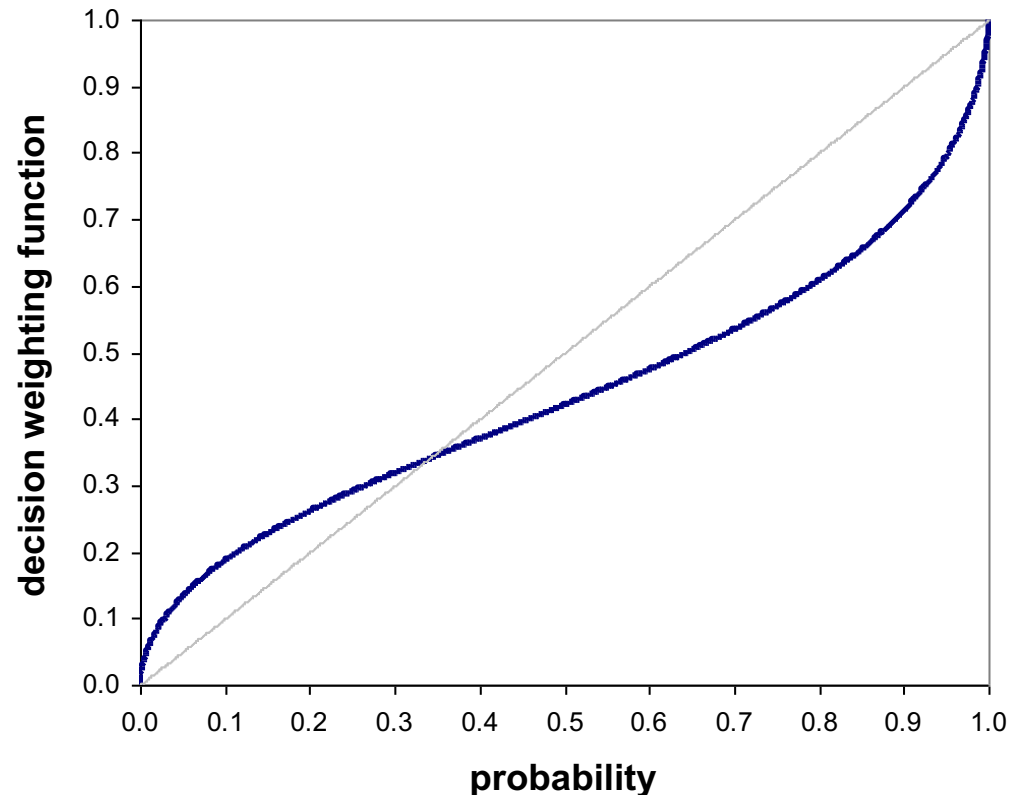
Risky Choice Observations III

- Effect of comparison (Fox & Tversky, 1995)
 - Gambling scenario without social comparison
 - Kaufman Broad Homes is one of the largest home sellers in America. Their stock is traded on the New York Stock Exchange.
 - Do you think that KBH stock will close higher or lower on Monday?
 - Would you prefer:
 - Receive \$50 for sure
 - Receive \$150 if your prediction is correct
 - Gambling scenario with social comparison
 - We are presenting the survey to undergraduates at San Jose University, graduate students in economics at Stanford University, and to professional stock analysts.
 - Would you prefer:
 - Receive \$50 for sure
 - Receive \$150 if your prediction is correct

Decision Weighting Function

The decision weighting function:

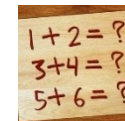
- is S-shaped (concave then convex)
- exhibits diminishing sensitivity (like the value function) away from the boundaries (0 and 1)
- illustrates
 - the certainty effect
 - the possibility effect



Identify some situations in which individuals seem to be overweighting small probabilities? What's the implication?

Mental Accounting

- Accounting (Random House College Dictionary):
 - “The theory or system of organizing, maintaining, and auditing the books of a firm.”
- Mental accounting consists of:
 - boundaries (budgets, accounts)
 - rules of combination
 - machinery of evaluation of results (prospect theory value function)
- 4 rules of combination (derived from hedonic maximization)
 - separate gains
 - integrate losses
 - integrate small losses with large gains
 - segregate small gains from large losses



Why does prospect theory predict these “rules”?

House Money Effect

- Effect of prior outcomes on choice (Thaler & Johnson, 1990)
 - You have just won \$30. Choose between:
 - 50% chance to gain \$9 and 50% chance to lose \$9 [70%]
 - no further gain or loss [30%]
- Big \$ example (Gertner, 1993; Card Sharks)
 - \$0.60 increase in bet with \$1.00 increase in cash stake
 - \$0.01 increase in bet with \$1.00 increase in non-cash winnings (e.g., cars)

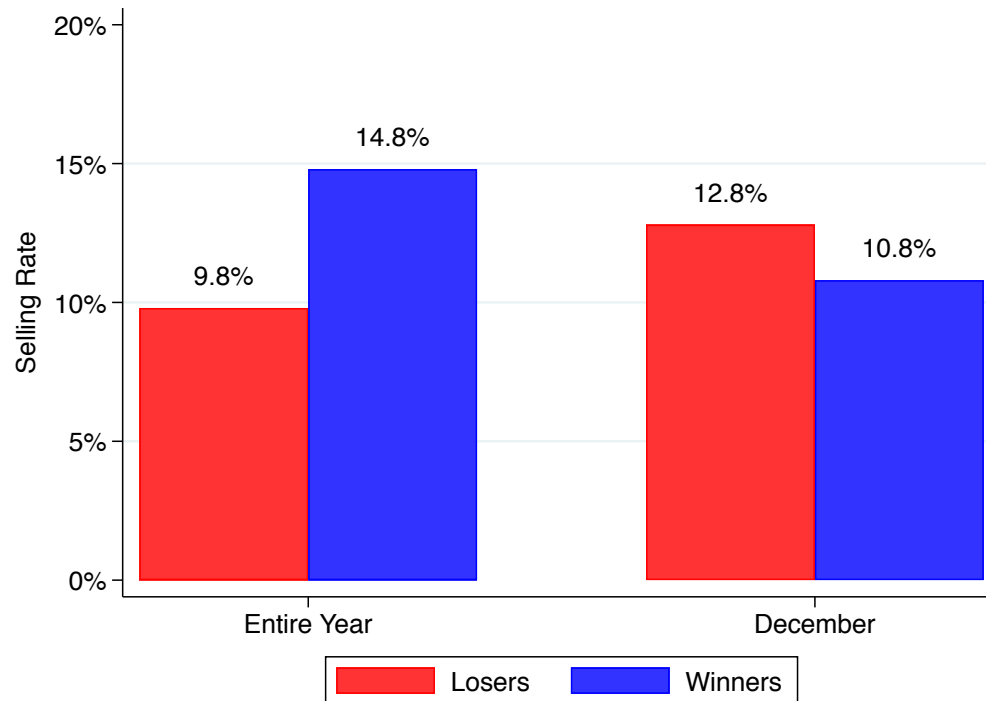
Trading and Mental Accounting

- Coval and Shumway (2005)
 - ❑ Study of CBOT traders
 - ❑ Risk as a function of morning profits
 - ❑ Risk-seeking in losses (for morning losers)
 - ❑ House money effect (for deep-in-the-money morning winners)

Other Related Phenomena I

1. Disposition Effect (Reluctance to sell losers)

- Study (Odean, 1997): 10,000 customer accounts of nationwide brokerage house (1987-1993); 162,948 records
- Result:



- Winning stocks outperform losing stocks (3.6% over 2 years)

Other Related Phenomena II

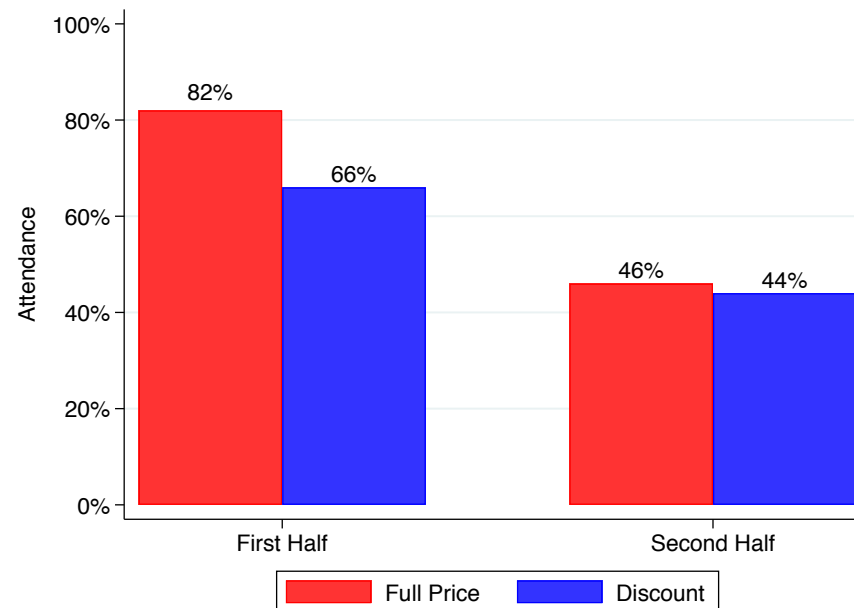
2. Equity Premium Puzzle

- Equity Premium: Return on stocks is about 6% higher than risk-free rate (over 70 years)
 - can't be explained by risk aversion:
 - 50% chance at \$100K, 50% chance at \$50K $\approx$ \$51.2K
 - Explanation (Benartzi & Thaler, 1995):
 - myopic loss aversion
 - annual evaluation

Other Related Phenomena III

3. Sunk Costs (Arkes and Blumer, 1985)

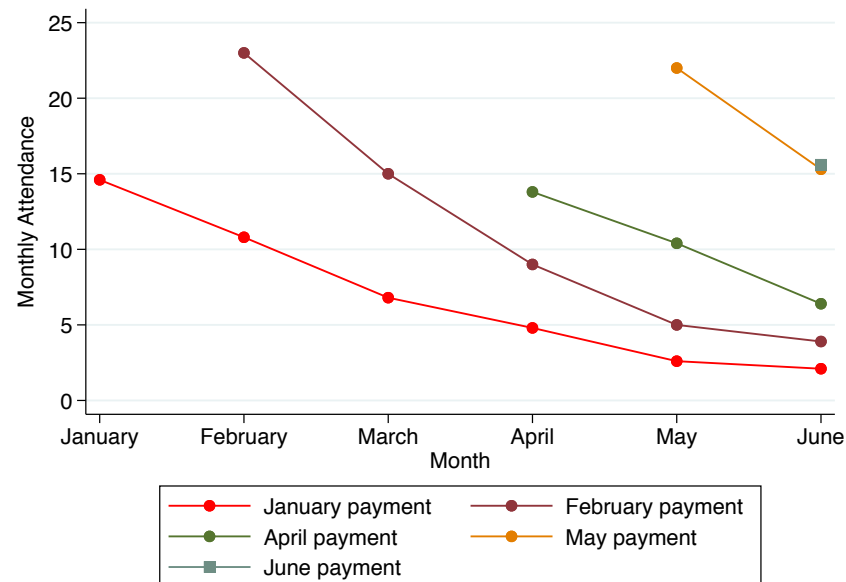
- Study: Subjects were people ready to buy season tickets for campus theatre group (Ohio University)
 - 1/3 paid full price
 - 1/3 unexpected small (13%) discount
 - 1/3 unexpected large (47%) discount



Other Related Phenomena V

3. Sunk Costs (Gourville and Soman, 1998)

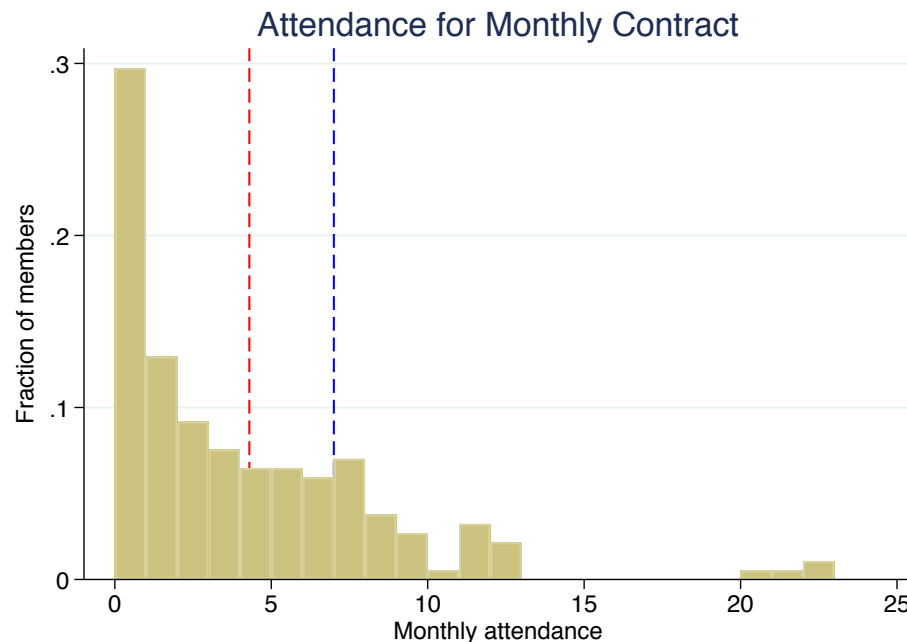
- Study: University of Colorado gym attendance for semi-annual members
 - Some members pay January 1
 - ...
 - Some members pay June 1



Other Related Phenomena V

4. Aggregation vs. Segregation (DellaVigna and Malmendier, 2006)

- Study: Gym attendance by contract
 - Monthly contract (\$70 to \$85)
 - Pay-per-visit (\$100/10 pass or \$12/visit)

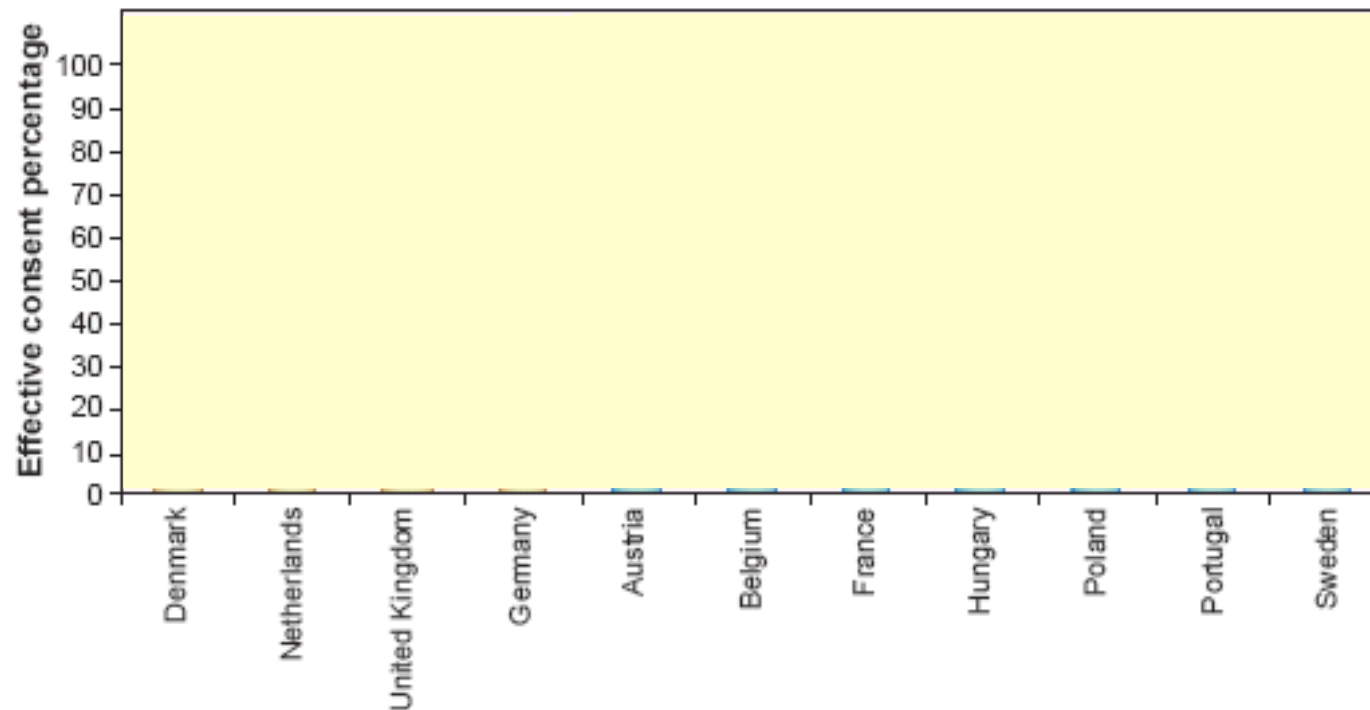


Average Cost per visit

- Monthly: \$17
- Pay-as-you go: \$10 or \$12

Other Related Phenomena VI

- Impact of defaults on organ donation (Johnson & Goldstein, 2003)



Effective consent rates, by country. Explicit consent (opt-in, gold) and presumed consent (opt-out, blue).

A Mini-Application

You have been hired as special consultant to the director of Human Resources at McKinsey (or some other firm that hires many MBA graduates). Based on the principles of mental accounting, how should McKinsey structure their compensation package to maximize the number of offers accepted (recruiting)? (Note: Of course, we are not looking for the following answer: “Increase salaries.” Consider the total pool of compensation fixed.) How should McKinsey structure compensation to retain employees? Again, take total compensation to be fixed. Does mental accounting suggest why different compensation packages might be required for recruiting and retention?

Regulating your level of risk taking

- How to become more or less risk averse?

Introduction to Behavioral Finance

- Modern financial economics (“Efficient Markets”) is based on two “rationality” assumptions
 - rational expectations (e.g., Bayesian updating)
 - expected utility maximization
- Both assumptions are descriptively inaccurate
 - Overconfidence/optimism
 - Prospect theory
- Behavioral Finance suggests (contrary to Efficient Markets) that
 - Security prices are highly correlated with intrinsic value, but sometimes diverge to a significant degree
 - It is possible to predict stock prices, but not with great precision

Selected Financial “Anomalies”

Efficient Markets Implications	Reality
Diversification	-Most investors are poorly diversified (“1/n heuristic”) -Home bias
Virtually no trading	-Heavy trading
Prices are not predictable	-Small firms, prior losers, low P/E, low price to book all outperform the index -Post-earnings price drifts
Small Risk Premium for Equity	-Equity Premium Puzzle

Diversification Heuristic

- Study 1 (Simonson, 1990): Choice of snacks
 - Simultaneous Choice: for each of the next three weeks
 - Sequential Choice: on each of the next three weeks
 - Results
 - Simultaneous: 64% choose three different snacks
 - Sequential: 9% choose three different snacks
- Study 2 (Read and Loewenstein, 1995): Trick-or-treaters
 - Simultaneous Choice: Choose two candies
 - Sequential Choice: Choose one of two candies
 - Results
 - Simultaneous: 52% choose different candies
 - Sequential: 0% choose different candies

Diversification Heuristic: Pensions I

- The “ $1/n$ heuristic” from 30,000 feet
 - Money Market Directories data base
 - 170 plans
 - 1.5 million participants
 - \$3.2 billion annual contributions
 - \$50 billion in assets
 - Number of equity funds
 - 62.2%
 - Money invested in equity
 - 61.8%

Diversification Heuristic: Pensions II

- “1/ n heuristic” continued
 - TIAA-CREF: The most popular asset allocation is 50-50 (bonds/equity)
 - Harry Markowitz (father of modern portfolio theory) on his TIAA-CREF choice:
 - “I should have computed the historic covariances of the asset classes and drawn an efficient frontier. Instead, ... I split my contributions fifty-fifty between bonds and equities.”
 - TWA pilots plan: 5 stock funds and 1 fixed income fund
 - 75% stocks
 - University of California plan: 1 stock fund and 4 fixed income funds
 - 34% stocks

Diversification Heuristic: Pensions III

- One implication...
 - Plans that do not offer company stock
 - average allocation is
 - 49% equity
 - 51% fixed income
 - Plans that offer company stock
 - the average allocation is:
 - 42% company stock
 - 29% other equities
 - 29% fixed income

Participation Rates in 401(k): Part I

- Vanguard Center for Retirement Research (Iyengar & Wei, 2004)
 - Data (2001)
 - 793,794 employees
 - 647 defined contribution plans
 - Plan options: 2 to 59 (median is 13)
 - Regression analysis to understand relationship between participation rates and number of plans
 - 2 options – 75%
 - 59 options – 61%

Participation Rates in 401(k): Part II

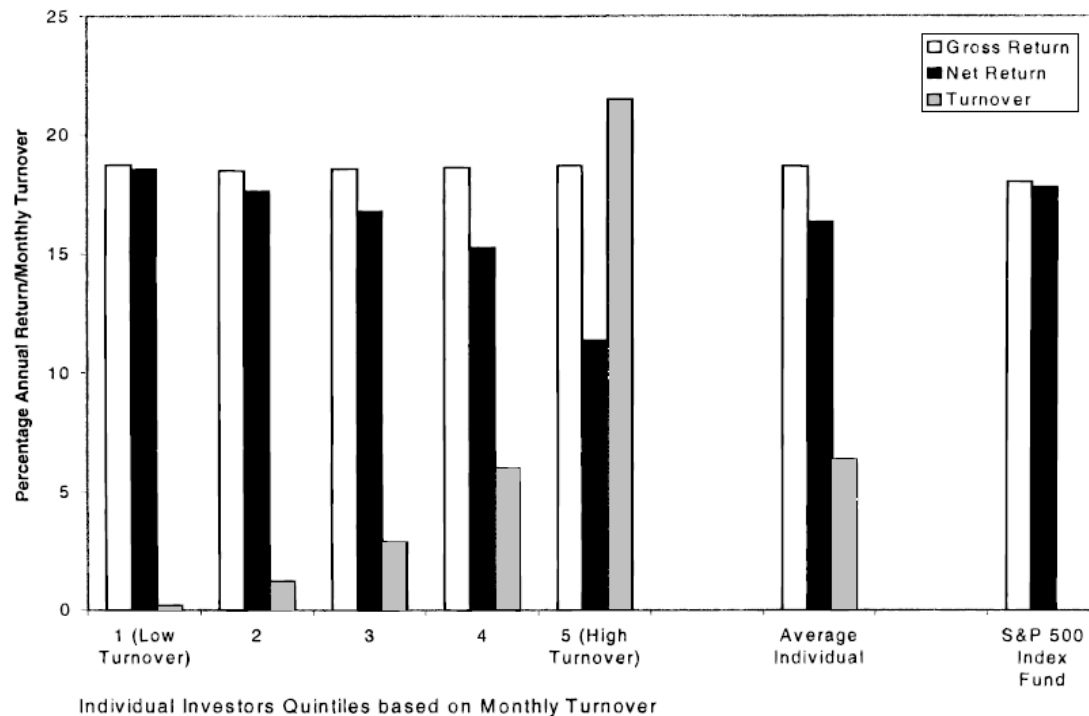
- Automatic Enrollment (Madrian & Shea, 2001)
 - Firm changes default: Participants are enrolled into the plan unless they explicitly opt out (a framing effect).
 - Result: Enrollments increase dramatically!
 - In one company, enrollments of new employees go from 49% to 86%.
 - Reasons: Status Quo Bias/Inertia
 - Many (most) employees have the same asset allocation as the day they started work.
 - Some married workers still list their mother as their beneficiary.
 - In one group of self-reported undersavers, 35% announced an intention to increase their savings rates within the next few months. But 86% of these ended up doing nothing.

Home Bias

- Investors tend to invest in familiar stocks
 - French & Poterba (1991): percentage of equities in domestic stocks
 - US: 94%
 - UK: 82%
 - Japan: 98%
 - Requires belief of 2% to 5% excess return
 - Benartzi (2001): investment in company stock
 - Over 30% of defined contribution plan assets are invested in employer stock
 - Huberman (2001): investment in local stock
 - Shareholders are much more likely to hold their regional “Baby Bell” than other “Baby Bells”

Excessive Trading

- Barber and Odean (2000)
 - Study of 66,545 households using a discount broker (1991-1996)



- Men (cost: 2.65%) trade 45% more often than women (cost: 1.72%)

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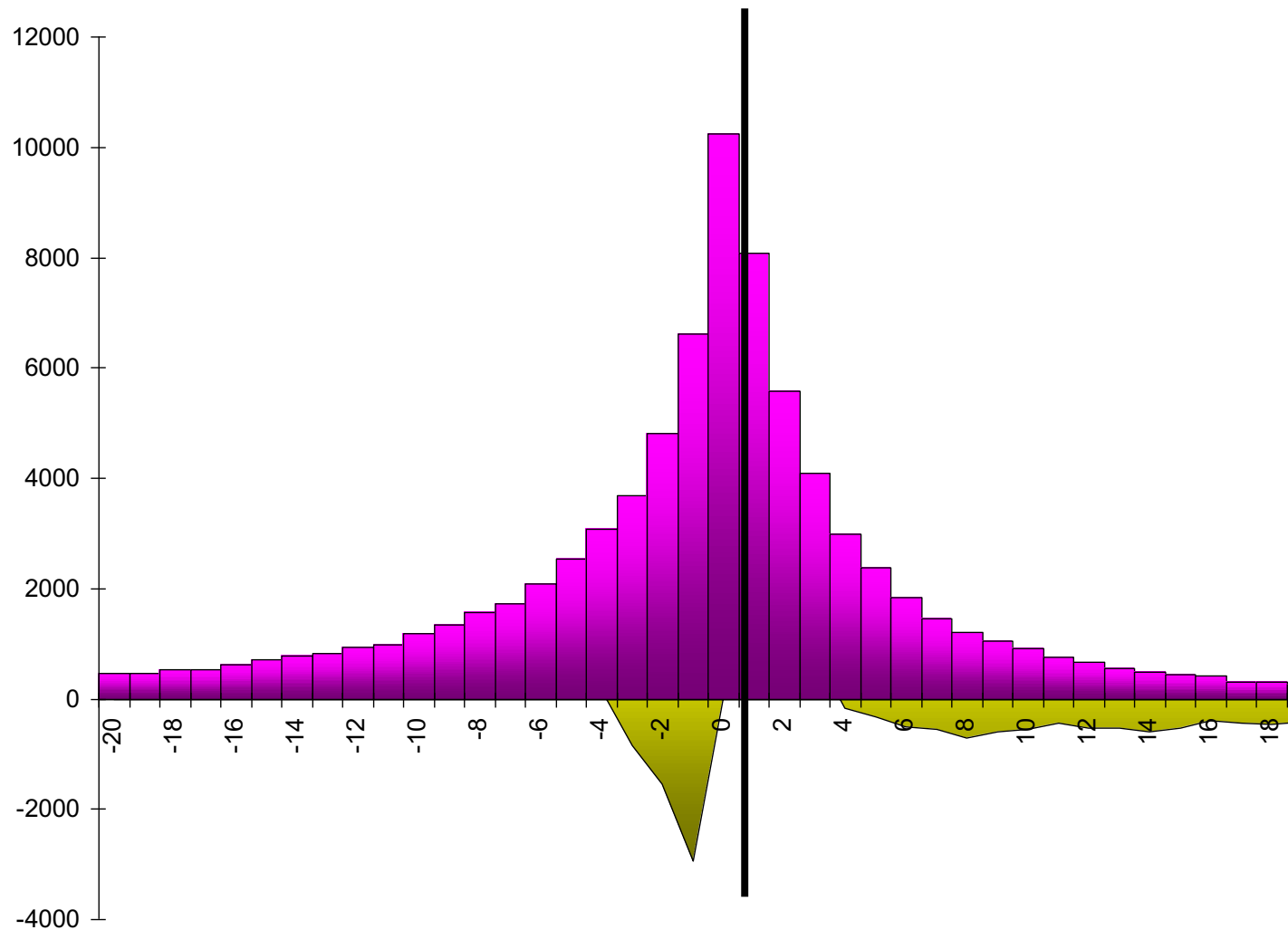
Under-reaction

- Jegadeesh and Titman (1993)
 - Stocks grouped by decile of 6 month returns
 - Top decile outperforms lowest decile by 10% on annual basis
- Under-reaction results (primarily) from two behavioral biases
 - Overconfidence
 - Individuals place too much confidence in their existing information and/or knowledge.
 - Anchoring
 - Individuals are tied to their previous view or opinion.

Earnings Manipulation

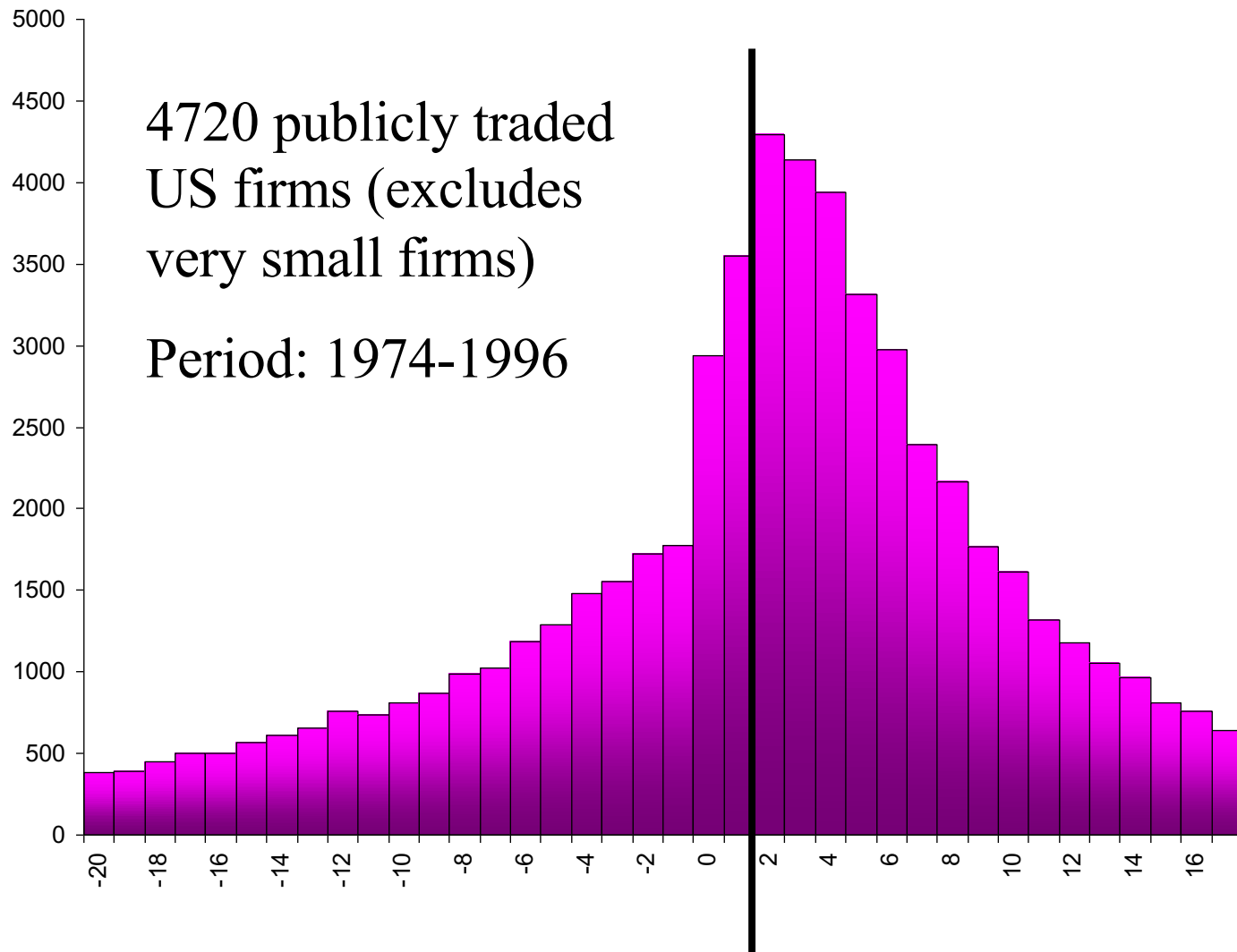
- Degeorge, Patel, & Zeckhauser (1999)
 - Three thresholds found to be empirically important:
 - Analysts' expectations: Beat forecast
 - Change in EPS: Sustain recent performance
 - EPS: Avoid red Ink
 - Findings
 - “True” earnings that just miss a threshold are manipulated to cross them
 - Firms that just cross thresholds do worse the next quarter

Forecast Error: “beat forecast”



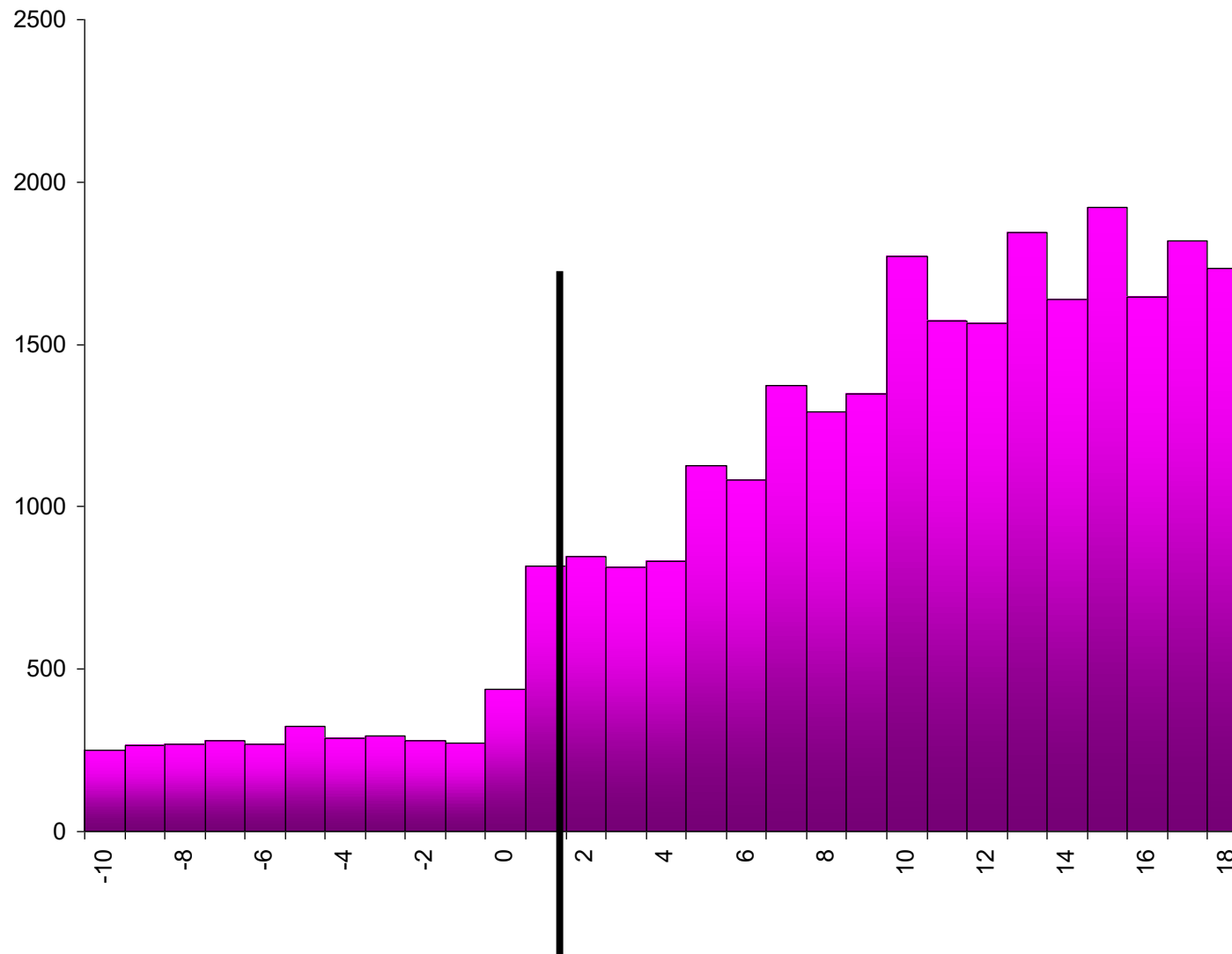
For Reference Only!

Change in EPS: “sustain performance”



For Reference Only!

EPS: “avoid red ink”



For Reference Only!

Stock Options and Risk Aversion

- Finding (Huddart and Lang, 1996)
 - Employees sacrifice 25% of an option's expected value by exercising (too?) early
- Empirical Study (Heath, Huddart, and Lang, 1999)
 - Study of 50,000 employees at 7 corporations over 9+ years
 - Probability of exercising stock option:
 - is positively correlated to recent stock returns (one month)
 - is negatively correlated to longer horizon stock returns
 - doubles when stock price exceeds maximum price over last year

Wu's Clues: Part 4



Clue 14: Don't let regret keep you or others from acting.

Clue 15: (Everything is relative.)

Clue 16: 1% is 1%.

Clue 17: Think about your decisions in terms of a *policy*, not one-at-a-time.
